

Synopsis

In this thesis, we have studied ternary spinodal decomposition in which a homogeneous and disordered ternary alloy, quenched into the ternary miscibility gap, produces a three-phase microstructure: A-rich α phase, B-rich β phase and C-rich γ phase.

An early work of de Fontaine analyzed the stability of a bulk ternary system with respect to composition fluctuations, in terms of the eigenvalues and eigenvectors of the stability matrix formed by the Jacobian of bulk free energy, f :

$$\begin{pmatrix} \partial^2 f / \partial c_A^2 & \partial^2 f / \partial c_A \partial c_B \\ \partial^2 f / \partial c_A \partial c_B & \partial^2 f / \partial c_B^2 \end{pmatrix}$$

where c_A and c_B are the mole fractions of A and B, respectively. The eigenvectors give the directions in the composition space along which the growth of composition fluctuations result in the steepest fall in the free energy of the system.

These results were confirmed by Chen and Eyre who used computer simulations to study spinodal decomposition in an idealized ternary system in which the three components are equivalent to one another (i.e., the interfacial energies are equal; $\sigma_{\alpha\beta} = \sigma_{\beta\gamma} = \sigma_{\alpha\gamma}$). However, it is important to note that for each spinodally decomposing alloy, there is a range of composition fluctuations which can grow. The choice of fluctuations that actually grow may be affected by several factors, including the interfacial energy and elastic strain energy associated with those fluctuations. With this motivation, we have studied

the effect of relative interfacial energies between the coexisting phases on the temporal evolution of microstructures produced by ternary spinodal decomposition.

In the present work, we have employed a diffuse-interface phase-field model to study ternary spinodal decomposition. In the model, the solid solution is treated as a continuous medium and the total free energy functional for the system is constructed in terms of the spatially varying composition fields using the Cahn-Hilliard (CH) formalism. The functional consists of two kinds of terms: a bulk free energy term and gradient energy terms. This model allows us to examine (by changing the gradient energy coefficients) the influence of interfacial energy on the morphology. The two dimensional computer simulations are based on a semi-implicit numerical integration of time-dependent CH equations in Fourier space.

System 1 is characterized by equal values of the interaction energies χ_{AB} , χ_{BC} and χ_{AC} (which determine the bulk thermodynamic behaviour of each of the three binary alloy systems); it is also characterized by equal values of the gradient energy coefficients, K_A , K_B and K_C (which determine the interfacial energies; $\sigma_{\alpha\beta} = \sigma_{\beta\gamma} = \sigma_{\alpha\gamma}$). Thus, it is an ideal system in which the three components are equivalent to one another. In our study, we use this system essentially for comparing our results from system 2 (with $\sigma_{\alpha\beta} = \sigma_{\beta\gamma} \sim 2.2\sigma_{\alpha\gamma}$) in which species B, since K_B is larger than $K_A = K_C$, is special (not equivalent to the other species).

In our detailed study of the five different alloys in the two systems, we have analyzed the interfacial energy effects in terms of microstructures, structure functions, kinetic paths and kinetics of coarsening. Our results indicate that the higher energy of the interfaces around the B-rich β phase has a different effect on two different categories of alloys: those

(with $c_B < 0.4$) in which initial decomposition does not involve partitioning of species B , and those (with $c_B > 0.4$) in which it does. The interfacial energy effects are much stronger in the latter category of alloys than those in the former. For each category, the broad conclusions are listed as follows:

Category I: Alloys with $c_B < 0.4$ (P_2 , Q_2 and T_2):

In these alloys, early stage decomposition is characterized by a lack of partitioning of species B . Hence, the alloys undergo a pseudo-binary phase separation leading to the formation of A -rich and C -rich regions. The length scale of the initially decomposed domains in system 2 is larger than that in system 1 by about 10-15 %. Partitioning of B begins only during intermediate stages, and results in the formation of coarse β particles in the regions between the α and γ particles. The β particles in system 2 are larger, and coarsen faster, than those in system 1.

Category II: Alloys with $c_B > 0.4$ (R_2 and S_2):

In these alloys, partitioning of species B is essential right from the beginning of decomposition, and creates coarse B -rich and B -poor regions; due to the larger length scale of these microstructures, which are about 40-50% larger than that in system 1, the initial decomposition is also much more sluggish than in system 1. This sluggishness is also evident in the time taken for the microstructure to exhibit all the three phases ($t \sim 1000$ in system 2, as opposed to $t \sim 500$ in system 1). Thus, the effect of interfacial energy is much stronger in this category of alloys than those in category I.

In all the alloys, the microstructures are coarser in system 2 than in system 1. This coarseness is more pronounced for the β particles than for the α and γ particles. Since the β phase has higher energy interfaces than the other two phases, its coarsening rate is

much higher in system 2 than in system 1. However, there is not much difference in the coarsening rates of the other two phases across the two systems.

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Chapter 1

Introduction

Metallurgy, as Prof. Peter Haasen observed in his classic textbook “Physical Metallurgy” [1], is a subject which *relates the properties of metals and metallic ‘mixtures’ or alloys to their microstructures*. Many technologically important alloys are multicomponent systems, exhibiting multiple phases in their microstructures. More importantly, one or more of these phases may also be formed as a result of phase transformations induced during processing. Since performance of these multicomponent alloys depends crucially on the morphology of the phases, there is a strong need for a fundamental understanding of the microstructures and of the kinetics of their evolution. This thesis attempts to address this need in the context of spinodal decomposition in ternary alloys which will lead to three-phase microstructures.

The phase separation dynamics in *binary alloys* undergoing spinodal decomposition have been extensively studied, both experimentally and theoretically, and early investigations were summarized by Hilliard [2]; Gunton *et al.* [3] and Binder [4] have reviewed the recent developments. But there are only a few studies on the thermodynamics and

kinetics of phase separation in *ternary* systems.

Ternary phase diagrams for simple model systems were calculated by Meijering [5, 6] and Kikuchi *et al.*, [7] and Kikuchi [8]. The thermodynamic stability and the early stages of spinodal decomposition kinetics for ternary systems were analyzed by de Fontaine [9, 10] and Morral and Cahn [11]. Hoyt [12] extended Langer's theory [13] for binary solutions to the case of ternary alloys and derived nonlinear diffusion equations for the local composition and three independent partial structure functions.

The nonlinear dynamics of the microstructural evolution during the process of spinodal decomposition in ternary alloys have been studied by Chen [14] and Eyre [15] using computer simulations. Their results on the early stages of decomposition confirm the results of the stability analysis of de Fontaine [9, 10]. As we show in chapter 4, this analysis identifies composition fluctuations which may grow and those which cannot, depending only on the alloy composition. However, it is important to note that for each alloy in both global and conditional spinodal regions of the phase diagram, there is a range of composition fluctuations which can grow. The choice of fluctuations that the system prefers to let grow may be affected by several factors, including the interfacial energy and elastic strain energy associated with those fluctuations. In this thesis, we examine the influence of interfacial energy on microstructural evolution and its kinetics during ternary spinodal decomposition.

We have employed a diffuse-interface phase-field model to study the microstructural evolution when a homogeneous, disordered ternary alloy is quenched into a ternary miscibility gap. In the model, the solid solution is treated as a continuous medium and the total free energy functional for the system is constructed in terms of the spatially varying

composition fields using the Cahn-Hilliard (CH) formalism [16]. The functional consists of two kinds of terms: the bulk free energy which is given by the nearest neighbour bond model using mean field approximation, and gradient energies that are given by the square of the gradients of the composition fields. The interfacial energy between the coexisting phases can be varied systematically by varying the gradient energy coefficients without changing the bulk thermodynamic behaviour of the system. The two dimensional computer simulations are based on a semi-implicit numerical integration of time-dependent CH equations in Fourier space.

In chapter 2 of the thesis we summarise the important experimental and theoretical studies on phase separation in binary and ternary alloys. In chapter 3, the model and the technique for solving the diffusion equations are described. In chapter 4, we describe the microstructures obtained by computer simulations and discuss the effect of interfacial energies. In chapter 5 we present a summary of the key conclusions from our study.

Chapter 2

Literature Review

Due to its fundamental importance, spinodal decomposition has been studied extensively using a variety of experimental and theoretical tools. These studies span both binary and multicomponent systems. In this chapter, we give a brief overview of the literature with a special emphasis on ternary phase separation.

2.1 Experiments

There have been a large number of experimental studies devoted to spinodal decomposition in binary systems. One of the earliest experiments was done by Rundman and Hilliard [17] who performed small angle X-ray scattering experiments to study the early stages of spinodal decomposition in Al-Zn alloy. They observed that the changes in the small-angle scattering pattern agree with Cahn's linearized theory [18] of the early stages of spinodal decomposition. Detailed reviews of various aspects of experimental investigations on spinodal decomposition in binary systems are given by Hilliard [2],

de Fontaine [19], Gunton *et al.*, [3] and Binder [4]. However, very few experimental investigations have been reported concerning spinodal decomposition in ternary systems. Ternary alloys such as Cu-Ni-Fe, Cu-Ni-Cr, Al-Ni-Co have been studied by various groups [20–22]. These studies reveal that these alloys undergo phase separation leading to only two phase microstructures.

Some ternary alloys (e.g., Al-Li-Zr, Inconel, alloy steels) yield microstructures comprising of three distinct phases though the mode of transformation in these alloys is different from spinodal [23–25]. Louisemakin and Ralph [23] have used electron microscopy to study the aging behaviour of an Al-rich Al-Li-Zr alloy over the temperature range 433 to 533K. A dispersion of Al_3Zr precipitates were present before the aging treatment. They have reported that the aging treatment produces a second phase, $\text{Al}_3\text{Li}(\delta')$ at the Al_3Zr /matrix interface which coarsens and completely envelopes the Al_3Zr particles. Another interesting example of two-phase precipitation is provided by the high temperature superalloy, Inconel. He *et al.* [25], have studied the interfaces in modified Inconel 718 alloys containing compact precipitates using high resolution electron microscopy. These precipitates were confirmed to be cube-shaped γ' precipitates, all the six faces of which are coated with three variants of disc-shaped γ'' precipitates. Shibita *et al.* [26], also reported compact precipitates of γ'' on γ' in Alloy 706.

To our knowledge, concrete experimental evidence of ternary spinodal decomposition in metallic alloy systems is not available till date. However, a few ternary polymer blends do exhibit ternary spinodal decomposition. Recently, Cavanaugh *et al.* [27] have studied multiphase polymer blends. They have used compositional quenching technique to prepare these polymer blends: two or more incompatible polymers are dissolved in a common solvent to prepare a homogeneous mixture; then the solvent is rapidly removed

by flash devolatilization. Thus, the initial, single phase mixture is plunged deeply into the multiphase region in which phase separation occurs via the spinodal mode. They have used optical phase contrast microscopy, SEM, TEM and HVEM to characterize the morphology of these phase separated blends. They have investigated two different polymeric systems: (a) PMMA (polymethyl methacrylate), PS (polystyrene), and PB (polybutadiene), and (b) PP (polypropylene), PS, and PB. Both of these blends are found to decompose via spinodal mode producing a three-phase blend morphology.

Several other studies have also examined phase separation in ternary polymer blends [28, 29], using light scattering as the main experimental tool. Whether the spontaneous decomposition in these experiments produces two-phase or three-phase microstructures can be established only with additional experiments. Thus, the extensive study of Cavanaugh *et al.* [27], is the only one (to our knowledge) which clearly establishes ternary spinodal decomposition in certain polymer blends.

2.2 Theoretical Investigations

Most of the early theoretical works on ternary systems were mostly concerned with thermodynamic stability analysis and construction of phase diagrams in model ternary systems. Kikuchi developed a method for efficient calculation of phase diagrams in model ternary systems [8]. Kikuchi *et al.* [7], have also computed isothermal sections of several typical phase diagrams for clustering and ordering ternary fcc solid solutions.

Morral and Cahn [11] extended Cahn's theory of spinodal decomposition [18] for binary systems to ternary systems. They analyzed the early stage spinodal decomposition

kinetics in ternary systems using the extended version of the linearized kinetic equations. De Fontaine [9, 10] analyzed the stability of multicomponent solid solutions with respect to diffusional processes as well as ordering. He employed a microscopic, or lattice, model of multicomponent solid solutions for which the free energy can be written using a mean field approximation. The stability of a homogeneous alloy was discussed in terms of eigenvalues and eigenvectors of the matrix of the free energy quadratic.

Hoyt [12] extended Langer's theory [30] of binary spinodal decomposition to the case of ternary alloys. He developed a ternary master equation and derived the equations of motion for the three independent partial structure functions. In a later paper [31], he discussed the solutions for the three partial structure functions in ternary systems at the early stages of decomposition by employing different values for the gradient energy coefficients, atomic mobilities and average compositions.

The nonlinear dynamics of inhomogeneous morphological evolution during spinodal decomposition in ternary alloys was studied by Chen using computer simulation techniques [14, 32]. He studied the sequence of phase separation in ternary systems for various alloy compositions and justified his results using de Fontaine's thermodynamic stability analysis. Recently, Eyre [15] has also studied the sequence of decomposition in ternary alloys using a diffuse interface model. But, both Chen and Eyre have taken the relative interfacial energies between the coexisting phases to be equal.

Huang *et al.* [33], has studied the interfacial segregation of a minority component at the interfaces between two phases in a model ternary alloy system. They also studied the effect of the gradient energy coefficients on the interface composition profiles. They observed that the gradient energy coefficient of the minority component has a very strong

influence on the extent of segregation of that component.

De Fontaine's analysis [34] considers only the bulk free energy. It predicts a range of fluctuations which can grow. Since Chen [14, 32] and Eyre [15] considered systems in which $\alpha\beta$, $\beta\gamma$ and $\alpha\gamma$ interfaces exhibit identical behaviour, their studies confirm the results from de Fontaine's stability analysis [34]. Since interfacial energy and elastic strain energy can play a significant role in deciding which fluctuation will eventually grow, we undertake a study of the effect of the former on microstructural evolution. Elastic stress effects are beyond the scope of this study: however, it is possible to incorporate them into our model.

Chapter 3

Formulation

In this chapter, we present our formulation for studying the dynamics of spinodal decomposition in a ternary system. We begin with an expression for the total free energy of a compositionally inhomogeneous ternary alloy using an extended form of the Cahn-Hilliard free energy functional [16]. Then, we present the kinetic equations that govern the temporal evolution of the composition fields; our derivation follows closely the work of Huang [33] on ternary polymer blends. In section 3.3, we present the details of the numerical technique used for solving the evolution equations.

In what follows, scalars are denoted by lightface letters, while vectors and matrices are denoted by boldface letters. Further, all the quantities in our formulation have been non-dimensionalized using a characteristic length l^* , a characteristic energy e^* and a characteristic time t^* . Some of the parameters may assume a value of unity during non-dimensionalization of the governing equations. However, for clarity, we present all the equations in our formulation without leaving these parameters out; thus, the resulting equations are also sensible in their dimensional form. A convenient choice of l^* , e^* and

t^* is presented at the end of this chapter.

3.1 Model system

We consider a ternary substitutional alloy consisting of the three atomic species say, A, B and C. Let $c_i(\mathbf{r}, t)$ ($i = A, B, C$) represent the mole fraction of the i 'th species as a function of position $\mathbf{r}$ and time t . Since c_i 's are the mole fractions, we have the following conditions:

$$\sum c_i(\mathbf{r}, t) = 1, \quad c_i(\mathbf{r}, t) \geq 0. \quad (3.1)$$

We assume that the bulk chemical free energy per atom, $f(c_A, c_B, c_C)$, of the homogeneous alloy is given by the regular solution expression:

$$f(c_A, c_B, c_C) = \frac{1}{2} \sum_{i \neq j} \chi_{ij} c_i c_j + \sum_i c_i \ln c_i, \quad (3.2)$$

where $i, j = A, B, C$. $\chi_{ij} = \chi_{ji}$ is the effective interaction energy between components i and j . For example, for the A-B solid solution, the effective interaction energy, χ_{AB} , is given by:

$$\chi_{AB} = \frac{Z[2E_{AB} - E_{AA} - E_{BB}]}{2k_B T}, \quad (3.3)$$

where E_{AB} , E_{AA} and E_{BB} are the bond energies between AB, AA and BB atoms, respectively, Z is the number of bonds per atom, k_B is the Boltzmann constant and T is the absolute temperature. If $\chi_{AB} > 0$, the binary system A-B exhibits a miscibility gap (phase separation) at low temperatures. On the other hand, if $\chi_{AB} < 0$ then the system exhibits ordering. χ_{BC} and χ_{AC} are defined in a similar manner.

For $\chi_{AB} = \chi_{BC} = \chi_{AC} = \chi \geq 3$, it can be shown [35] that the bulk free energy density has three minima and one maximum; in other words, the system exhibits a ternary mis-

cibility gap. At high temperatures, any alloy whose composition falls within this gap, is homogeneous and disordered. When such a single phase alloy is quenched to low temperatures ($\chi \geq 3$), the alloy may become unstable with respect to ternary phase separation.

Defining $f(c_A, c_B, c_C)$ as the bulk free energy per atom, the total Helmholtz free energy, $\mathcal{F}$, for an isotropic, three-component system with compositional inhomogeneity, is expressed in terms of the following Cahn-Hilliard free energy functional [16]:

$$\mathcal{F} = N_v \int_V \left[f(c_A, c_B, c_C) + \sum_i K_i (\nabla c_i)^2 \right] dV, \quad (3.4)$$

where $i = A, B, C$; N_v is the number of atoms per unit volume (assumed to be independent of composition and position) and K_A , K_B and K_C are the bare gradient energy coefficients associated with gradients of compositions of A, B and C, respectively.

3.2 Evolution equations for the composition field

Let $\mathbf{J}_i$ denote the diffusion flux of the i 'th component ($i = A, B, C$) relative to a lattice plane. With the assumption that the flux of the individual species is a linear and homogeneous function of the gradient in its chemical potential μ_i , the phenomenological flux equations with respect to a local (lattice) frame of reference can be written as

$$\mathbf{J}_i = -M_i \nabla \mu_i, \quad (3.5)$$

where M_i is the 'lattice reference frame' mobility of the i 'th component and is always positive. It should be noted that the mobility is a second rank property tensor. Since mobility is isotropic for cubic materials, it may be replaced by a scalar, as in Eqn. (3.5).

Since we consider substitutional alloys, some mechanism, say, vacancy mechanism, might be operative during diffusion such that the sum of fluxes, $\mathbf{J}_i$ across a given lattice

plane need not become zero. So, it is convenient to choose a stationary frame of reference across which the sum of the net fluxes of the three species vanishes; generally, one of the fixed ends of the specimen is chosen as the frame of reference ('laboratory reference frame') (see Shewmon [36] for details). The net flux of each component relative to laboratory reference frame $\bar{\mathbf{J}}_i$, is given by [2, 36]

$$\bar{\mathbf{J}}_i = \mathbf{J}_i - c_i(\mathbf{J}_A + \mathbf{J}_B + \mathbf{J}_C); \quad i = A, B, C. \quad (3.6)$$

From this equation [Eqn. (3.6)] and from Eqn. (3.1), it is clear that

$$\bar{\mathbf{J}}_A + \bar{\mathbf{J}}_B + \bar{\mathbf{J}}_C = \mathbf{0}. \quad (3.7)$$

Using Eqns. (3.5) and (3.6) the net flux of each of the components A, B and C can be written as

$$\begin{aligned} \bar{\mathbf{J}}_A &= -(1 - c_A)M_A \nabla \mu_A - c_A M_B \nabla \mu_B - c_A M_C \nabla \mu_C, \\ \bar{\mathbf{J}}_B &= -(1 - c_B)M_B \nabla \mu_B - c_B M_A \nabla \mu_A - c_B M_C \nabla \mu_C, \\ \bar{\mathbf{J}}_C &= -(1 - c_C)M_C \nabla \mu_C - c_C M_A \nabla \mu_A - c_C M_B \nabla \mu_B. \end{aligned} \quad (3.8)$$

Due to Eqn. (3.7), the expression for J_C is redundant. Further, from Eqn. (3.1), we conclude that we need to solve for the evolution of only two composition variables, say, c_A and c_B . Now, let us derive an expression for $\bar{\mathbf{J}}_A$ by applying the Gibbs-Duhem relation locally:

$$c_A \nabla \mu_A + c_B \nabla \mu_B + c_C \nabla \mu_C = \mathbf{0}; \quad i = A, B, C. \quad (3.9)$$

Using Eqns. (3.1) and (3.9) we get

$$\nabla \mu_C = -c_A \nabla \mu_A^{eff} - c_B \nabla \mu_B^{eff}, \quad (3.10)$$

where we have defined $\mu_A^{eff} = (\mu_A - \mu_C)$ and $\mu_B^{eff} = (\mu_B - \mu_C)$. Similarly, for $\nabla\mu_A$ and $\nabla\mu_B$ we can write

$$\nabla\mu_A = (1 - c_A)\nabla\mu_A^{eff} - c_B\nabla\mu_B^{eff}, \quad (3.11)$$

$$\nabla\mu_B = (1 - c_B)\nabla\mu_B^{eff} - c_A\nabla\mu_A^{eff}. \quad (3.12)$$

Using the expressions for $\nabla\mu_A$, $\nabla\mu_B$ and $\nabla\mu_C$ in Eqn. (3.8) and rearranging, we obtain

$$\begin{aligned} \bar{\mathbf{J}}_A = & -\{(1 - c_A)^2 M_A + (c_A)^2 M_B + (c_A)^2 M_C\} \nabla\mu_A^{eff} \\ & + \{c_B(1 - c_A)M_A + c_A(1 - c_B)M_B - c_A c_B M_C\} \nabla\mu_B^{eff}. \end{aligned} \quad (3.13)$$

For the component B, we can derive a similar expression:

$$\begin{aligned} \bar{\mathbf{J}}_B = & -\{(1 - c_B)^2 M_B + (c_B)^2 M_A + (c_B)^2 M_C\} \nabla\mu_B^{eff} \\ & + \{c_B(1 - c_A)M_A + c_A(1 - c_B)M_B - c_A c_B M_C\} \nabla\mu_A^{eff}. \end{aligned} \quad (3.14)$$

Now, let us define the effective mobilities M_{AA} , M_{BB} , and M_{AB} as follows:

$$M_{AA} = (1 - c_A)^2 M_A + c_A^2 (M_B + M_C), \quad (3.15)$$

$$M_{BB} = (1 - c_B)^2 M_B + c_B^2 (M_A + M_C), \quad (3.16)$$

$$M_{AB} = M_{BA} = -c_B(1 - c_A)M_A - c_A(1 - c_B)M_B + c_A c_B M_C. \quad (3.17)$$

Using the relations in Eqn. (3.15), we rewrite the flux equations 3.13 and 3.14 in a more compact form as follows:

$$\begin{aligned} \bar{\mathbf{J}}_A = & -M_{AA}\nabla\mu_A^{eff} - M_{AB}\nabla\mu_B^{eff}, \\ \bar{\mathbf{J}}_B = & -M_{BA}\nabla\mu_A^{eff} - M_{BB}\nabla\mu_B^{eff}. \end{aligned} \quad (3.18)$$

Following Cahn [37], the quantity $\mu_A^{eff} = \mu_A - \mu_C$ is the change in free energy when unit amount of A atoms is added ($\Delta\mathcal{F} = +\mu_A$) and an equal amount of C atoms is removed

($\Delta\mathcal{F} = -\mu_C$) reversibly. Similarly we define $\mu_B^{eff} = \mu_B - \mu_C$. From thermodynamics of homogeneous (bulk) phases, it can be easily shown that

$$\mu_A^{eff} = \frac{\partial f(c_A, c_B)}{\partial c_A}. \quad (3.19)$$

Extending this definition to our model of inhomogeneous systems, we define μ_i^{eff} as the variational derivative of $\mathcal{F}$ with respect to c_i , the total free energy of the system given by Eqn. (3.4):

$$\mu_i^{eff} = \frac{\delta \mathcal{F}}{\delta c_i}; \quad i = A, B. \quad (3.20)$$

From Eqn.(3.4), we obtain the following expressions for μ_i^{eff} :

$$\mu_A^{eff} = \partial f / \partial c_A - 2(K_A + K_C)\nabla^2 c_A - 2K_C\nabla^2 c_B, \quad (3.21)$$

$$\mu_B^{eff} = \partial f / \partial c_B - 2(K_B + K_C)\nabla^2 c_B - 2K_C\nabla^2 c_A, \quad (3.22)$$

where, $\partial f / \partial c_A$ and $\partial f / \partial c_B$ are given by:

$$\partial f / \partial c_A = \ln c_A - \ln c_C + (\chi_{AB} - \chi_{BC})c_B + \chi_{AC}(c_C - c_A), \quad (3.23)$$

$$\partial f / \partial c_B = \ln c_B - \ln c_C + (\chi_{AB} - \chi_{AC})c_A + \chi_{BC}(c_C - c_B). \quad (3.24)$$

The effective mobilities for a three-component system are represented in the form of a matrix as

$$\mathbf{M} = \begin{bmatrix} M_{AA} & M_{AB} \\ M_{BA} & M_{BB} \end{bmatrix} \quad (3.25)$$

According to Onsager's reciprocity theorem, $\mathbf{M}$ is symmetric i.e. $M_{ij} = M_{ji}$. Since, by the second law of thermodynamics, the rate of entropy production should always be positive for irreversible processes, using the second law and Onsager's principle, it can be shown [38] that $\mathbf{M}$ is positive definite i.e. $M_{AA}M_{BB} - M_{AB}^2 > 0$

The temporal evolution of the composition fields is described by the continuity equations:

$$\frac{\partial c_i}{\partial t} = -\nabla \cdot \bar{\mathbf{J}}_i; \quad i = A, B. \quad (3.26)$$

Using the Eqns. (3.18) and (3.21) in Eqn. (3.26) we get the following two independent kinetic equations for our model system:

$$\begin{aligned} \frac{\partial c_A}{\partial t} = & M_{AA} \{ \nabla^2 (\partial f / \partial c_A) - 2K_{AA} \nabla^2 c_A - 2K_{AB} \nabla^2 c_B \} \\ & + M_{AB} \{ \nabla^2 (\partial f / \partial c_B) - 2K_{AB} \nabla^2 c_A - 2K_{BB} \nabla^2 c_B \}. \end{aligned} \quad (3.27)$$

$$\begin{aligned} \frac{\partial c_B}{\partial t} = & M_{BA} \{ \nabla^2 (\partial f / \partial c_A) - 2K_{AA} \nabla^2 c_A - 2K_{AB} \nabla^2 c_B \} \\ & + M_{BB} \{ \nabla^2 (\partial f / \partial c_B) - 2K_{AB} \nabla^2 c_A - 2K_{BB} \nabla^2 c_B \}. \end{aligned} \quad (3.28)$$

For the sake of convenience, we denote $K_{AA} = K_A + K_C$, $K_{BB} = K_B + K_C$ and $K_{AB} = K_{BA} = K_C$. It should be noted that, the ‘laboratory reference frame’ mobilities in Eqn. (3.15) are assumed to be constants in our present work, though these are dependent on the composition fields.

3.3 Numerical implementation

We use a semi implicit Fourier spectral method, originally used for binary systems by Chen *et al.* [39], for solving the evolution equations. We have extended the technique to the ternary systems.

Fourier transform of the kinetic equations (3.27) and (3.28) transforms them into

a sequence of ordinary differential equations in Fourier space as follows:

$$\begin{aligned} \frac{\partial \tilde{c}_A(\mathbf{k}, t)}{\partial t} = & M_{AA} \{ -k^2 (\partial f / \partial c_A)_{\mathbf{k}} - 2K_{AA} k^4 \tilde{c}_A - 2K_{AB} k^4 \tilde{c}_B \} \\ & + M_{AB} \{ -k^2 (\partial f / \partial c_B)_{\mathbf{k}} - 2K_{AB} k^4 \tilde{c}_A - 2K_{BB} k^4 \tilde{c}_B \} , \end{aligned} \quad (3.29)$$

$$\begin{aligned} \frac{\partial \tilde{c}_B(\mathbf{k}, t)}{\partial t} = & M_{BA} \{ -k^2 (\partial f / \partial c_A)_{\mathbf{k}} - 2K_{AA} k^4 \tilde{c}_A - 2K_{AB} k^4 \tilde{c}_B \} \\ & + M_{BB} \{ -k^2 (\partial f / \partial c_B)_{\mathbf{k}} - 2K_{AB} k^4 \tilde{c}_A - 2K_{BB} k^4 \tilde{c}_B \} , \end{aligned} \quad (3.30)$$

where $\mathbf{k}$ is a vector in reciprocal (Fourier) space, and $k = |\mathbf{k}|$. $\tilde{c}_A(\mathbf{k}, t)$, $\tilde{c}_B(\mathbf{k}, t)$ are the Fourier transforms of the respective compositions in the real space. $(\partial f / \partial c_A)_{\mathbf{k}}$ and $(\partial f / \partial c_B)_{\mathbf{k}}$ are the Fourier transforms of the bulk driving force terms $\partial f / \partial c_A$ and $\partial f / \partial c_B$, respectively. Since our aim is to study the microstructural evolution in a ternary system, the computer simulations are performed by iterating Eqns. (3.29) and (3.30). Using backward differencing for $\partial \tilde{c}_A / \partial t$ and $\partial \tilde{c}_B / \partial t$, respectively, Eqns. (3.29) and (3.30) can be rewritten as:

$$\begin{aligned} \frac{\tilde{c}_A(\mathbf{k}, t + \Delta t) - \tilde{c}_A(\mathbf{k}, t)}{\Delta t} = & -k^2 [M_{AA} \tilde{g}_A(\mathbf{k}, t + \Delta t) + M_{AB} \tilde{g}_B(\mathbf{k}, t + \Delta t)] \\ & - 2k^4 [(M_{AA} K_{AA} + M_{AB} K_{AB}) \tilde{c}_A(\mathbf{k}, t + \Delta t) \\ & + (M_{AA} K_{AB} + M_{AB} K_{BB}) \tilde{c}_B(\mathbf{k}, t + \Delta t)] , \end{aligned} \quad (3.31)$$

$$\begin{aligned}
\frac{\tilde{c}_B(\mathbf{k}, t + \Delta t) - \tilde{c}_B(\mathbf{k}, t)}{\Delta t} = & -k^2 [M_{BA}\tilde{g}_A(\mathbf{k}, t + \Delta t) + M_{BB}\tilde{g}_B(\mathbf{k}, t + \Delta t)] \\
& - 2k^4 [(M_{BA}K_{AA} + M_{BB}K_{BA})\tilde{c}_A(\mathbf{k}, t + \Delta t) \\
& + (M_{BB}K_{BB} + M_{BA}K_{AB})\tilde{c}_B(\mathbf{k}, t + \Delta t)] ,
\end{aligned} \tag{3.32}$$

where we have defined $\tilde{g}_A = (\partial f / \partial c_A)_{\mathbf{k}}$ and $\tilde{g}_B = (\partial f / \partial c_B)_{\mathbf{k}}$.

The linear fourth-order terms, $\tilde{c}_A$ and $\tilde{c}_B$ in Eqns. (3.31) and (3.32), are treated implicitly while the non-linear terms, $\tilde{g}_A$ and $\tilde{g}_B$ are treated explicitly i.e., $\tilde{g}_A(\mathbf{k}, t + \Delta t) = \tilde{g}_A(\mathbf{k}, t)$ and $\tilde{g}_B(\mathbf{k}, t + \Delta t) = \tilde{g}_B(\mathbf{k}, t)$. This semi-implicit formulation yields the following equations which we solve iteratively for obtaining the microstructures:

$$\begin{bmatrix} Q_{AA} & Q_{AB} \\ Q_{BA} & Q_{BB} \end{bmatrix} \begin{bmatrix} \tilde{c}_A(t + \Delta t) \\ \tilde{c}_B(t + \Delta t) \end{bmatrix} = \begin{bmatrix} \tilde{c}_A(t) - k^2 \Delta t [M_{AA}\tilde{g}_A(t) + M_{AB}\tilde{g}_B(t)] \\ \tilde{c}_B(t) - k^2 \Delta t [M_{AB}\tilde{g}_A(t) + M_{BB}\tilde{g}_B(t)] \end{bmatrix} , \tag{3.33}$$

where,

$$\begin{aligned}
Q_{AA} &= 1 + 2(M_{AA}K_{AA} + M_{AB}K_{AB})k^4 \Delta t , \\
Q_{AB} &= 2(M_{AA}K_{AB} + M_{AB}K_{BB})k^4 \Delta t , \\
Q_{BA} &= 2(M_{BB}K_{AB} + M_{AB}K_{AA})k^4 \Delta t , \\
Q_{BB} &= 1 + 2(M_{BB}K_{BB} + M_{AB}K_{AB})k^4 \Delta t .
\end{aligned} \tag{3.34}$$

Discussions on the numerical technique

Chen [40] used the semi-implicit Fourier-spectral method to solve the CH equations for binary systems while Eyre [15] used the explicit Euler finite-difference scheme to solve

the CH equations for ternary systems. It has been observed that the explicit Euler scheme as well as the other higher order explicit finite difference schemes for CH equations have a time step constraint given by $\Delta t = (\Delta x)^4$. In case of binary systems, the semi-implicit Fourier-spectral method produces significantly more accurate results with less grid points and larger size of Δt .

For our technique, the value of $\Delta t = 0.05$ has been found to be numerically stable for the value of Δx lying between 0.5 and 2.0. The values of Δt greater than 0.05 produce numerically unstable results even in the case of 1-D. For checking the accuracy of the solutions we have doubled and halved Δx and calculated the profiles in 1-D keeping the system size same. We have found that the profiles lie exactly upon one another. A formal stability analysis of the scheme was not carried out.

3.4 Scaling of parameters

As mentioned in the beginning of the chapter, all the variables in the formulation are non-dimensionalized using an energy parameter e^* , a characteristic length l^* and a characteristic time t^* . We have chosen $e^* = k_B T$, $l^* = (K_A^*)^{1/2}/2$ and $t^* = (e^*/M_{AA}^* l^{*2})$ to render all the equations non-dimensional; the superscript ‘*’ indicates the dimensional quantities. In the subsequent chapters we present all our results in terms of the scaled parameters.

3.5 Scope and limitations of the model

Though our formulation is quite general, the present model does not include the effects of variable mobility and elastic stresses.

Elastic stress effects

The elastic stresses arise due to lattice parameter mismatch between the product and the parent phases. The contribution of elastic-strain energy to the total driving force for microstructural evolution can be incorporated using Khachaturyan's formalism where the strain energy is expressed as a functional of the composition fields by expressing the stress-free strain field in terms of $c_i(\mathbf{r}, t)$ assuming that the misfit strain is predominantly caused by the concentration heterogeneity. Thus, our model can be extended to study the elastic stress effects due to coherency strains for an elastically isotropic system by adding an extra term (the elastic energy) to the existing free energy functional.

Variable mobility

Though we have given the expressions for concentration dependent mobilities in our formulation, we have assumed the mobilities to be constant in the present simulations. Previously, Chen *et al.* [40] have solved the modified CH equation with variable mobility for a two-component system using the semi-implicit Fourier spectral method which is much more stable and accurate than the conventional explicit Euler finite difference scheme. But they have also shown that for binary systems the CH equation with variable mobility is computationally about 3 times slower than the CH equation with constant mobility because extra computational term is required to do the extra Fourier transforms in the former case. Though it would have been possible for us to use the concentration dependent mobilities in the existing model, we have used constant mobilities due to constraint in computational time.

Nucleation and growth

The alloys lying in the positive definite region (region *III*) (see Fig. 4.2) decompose via a process of nucleation and growth. For the B-rich and C-rich nuclei to form, sufficient energy in the form of thermal fluctuations should be introduced into the system to overcome the nucleation barrier. This process can be studied with our model by introducing sustained noise instead of an initial perturbation (used to study spinodal decomposition). The introduction of sustained noise ensure the formation of stable B-rich and C-rich nuclei.

Chapter 4

Results

In this chapter, we present our results for microstructural evolution in ternary systems.

The present work is aimed at understanding

- (a) the sequence of decomposition in ternary systems for different alloy compositions
- (b) the role of relative interfacial energies on the morphology.

These are studied by varying the compositions and the gradient energy parameters in a systematic manner, which in turn also affect the thermodynamics and kinetics of the system. In section 4.1, we briefly describe the stability of the ternary alloy with respect to spinodal decomposition for different alloy compositions. In section 4.2, we describe the two alloy systems that we have chosen for the simulations. In sections 4.4-4.7, we describe the microstructures and the kinetics of their evolution in these two systems for the different alloy compositions. In section 4.8, these results are discussed, to yield broad conclusions.

4.1 Thermodynamic Stability

Let us consider a symmetric A-B-C ternary system with $\chi_{AB} = \chi_{BC} = \chi_{AC} = \chi$. A value of $\chi = 3.5$ for the effective interaction energy parameter ensures that the system exhibits a ternary miscibility gap, schematically shown in the phase diagram in Fig. 4.1. In this diagram, the ‘A-rich’ phase is labeled as α , the ‘B-rich’ phase is labeled as β and the ‘C-rich’ phase is labeled as γ . At $\chi = 3.5$ the equilibrium compositions of α , β and γ phases in the three-phase alloy are $(c_A, c_B, c_C) = (0.91, 0.045, 0.045)$, $(0.045, 0.91, 0.045)$ and $(0.045, 0.045, 0.91)$, respectively.

The alloy compositions which we have chosen for our simulation are represented by the points P, Q, R, S and T in Fig. 4.1. The compositions represented by these points are listed in the table below (Table 4.1).

Table 4.1: Alloy compositions chosen for simulation

	c_A	c_B	c_C
P	1/3	1/3	1/3
Q	0.25	0.25	0.50
R	0.25	0.50	0.25
S	0.43	0.43	0.14
T	0.43	0.14	0.43

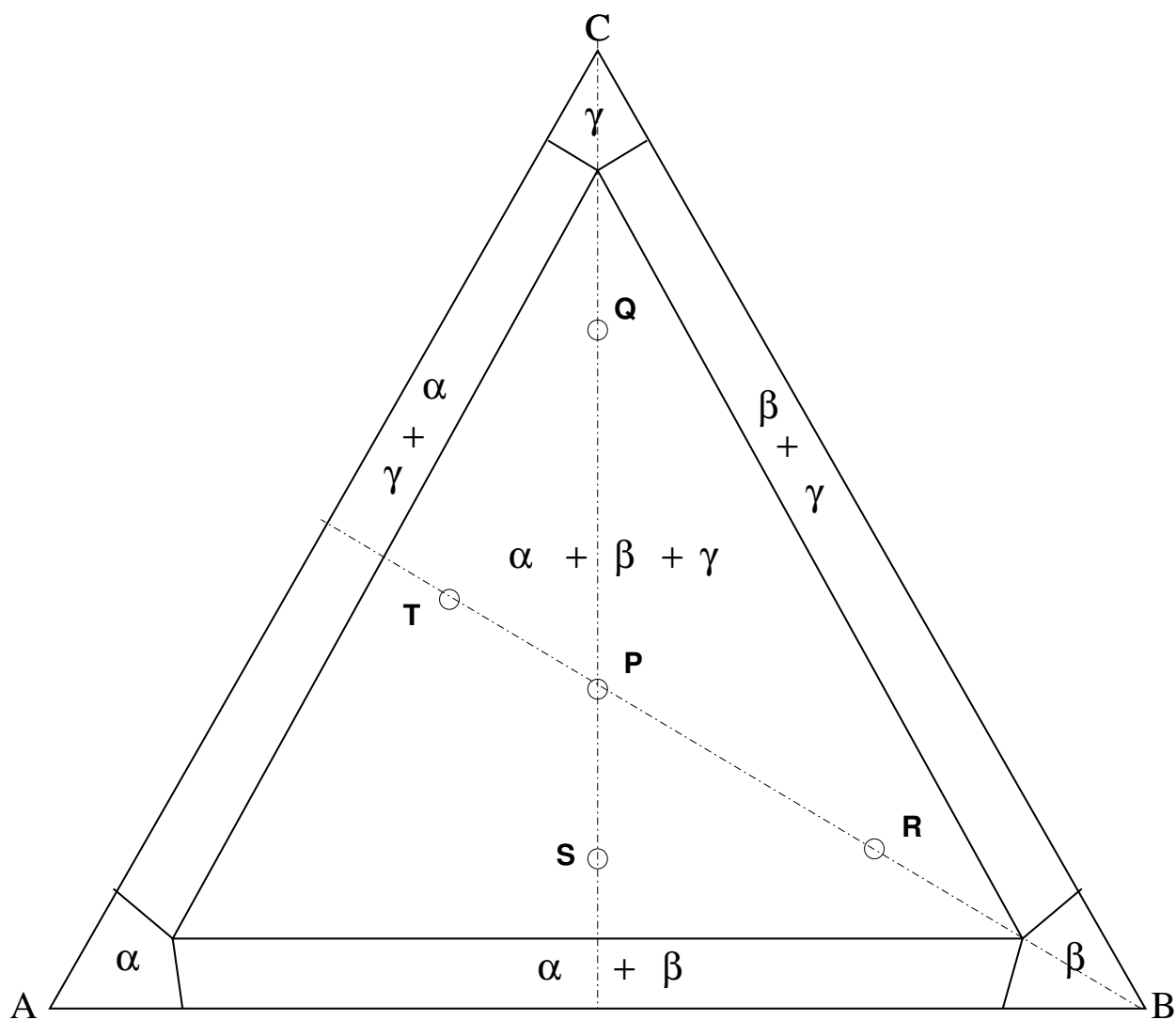


Figure 4.1: Isothermal section of the phase diagram for $\chi_{AB} = \chi_{BC} = \chi_{AC} = \chi = 3.5$ (schematic).

We now present the criterion for the stability of homogeneous alloys in this system with respect to infinitesimal composition fluctuations (i.e., with respect to spinodal decomposition). Our discussion is based on de Fontaine's analysis in terms of the eigenvalues and eigenvectors of the stability matrix formed by the second derivatives of the bulk free energy, f with respect to c_A and c_B [34]:

$$\begin{pmatrix} \partial^2 f / \partial c_A^2 & \partial^2 f / \partial c_A \partial c_B \\ \partial^2 f / \partial c_A \partial c_B & \partial^2 f / \partial c_B^2 \end{pmatrix}$$

Let us denote $f_{ij} = \partial^2 f / \partial c_i \partial c_j$ where, $i, j = A, B$. The expressions for f_{AA} , f_{BB} and f_{AB} for a ternary regular solution are obtained using Eqn. 3.2 as follows:

$$\begin{aligned} f_{AA} &= -2\chi + \left(\frac{1}{c_A} + \frac{1}{c_C} \right) \\ f_{AB} &= f_{BA} = -\chi + \frac{1}{c_C} \\ f_{BB} &= -2\chi + \left(\frac{1}{c_B} + \frac{1}{c_C} \right) \end{aligned}$$

where $c_C = 1 - c_A - c_B$.

In the region of absolute stability, where the stability matrix is positive definite (i.e., both its eigenvalues are positive), any arbitrary composition fluctuation imparts a positive sign to Δf , the difference in free energy between a solution containing composition fluctuations and a uniform solution which is devoid of any such fluctuation; thus, such fluctuations are unstable.

The region of absolute instability, characterized by a negative definite stability matrix whose eigenvalues are all negative, any arbitrary composition fluctuation continues to grow, since it reduces the system free energy. These two regions are separated by

some intermediate region where the solution is unstable with respect to certain composition fluctuations only. In these intermediate regions, the stability matrix is indefinite; one eigenvalue of the matrix is positive while the other is negative.

In general, for multicomponent systems comprising of n components, a $(n - 1)$ -dimensional surface in the n -dimensional space separates any two of the three regions described above. One of the n eigenvalues of the stability matrix vanishes at this surface, and has opposite signs on either side of the surface. This surface is referred to as the *spinodal* surface of the n -component system. Since, eigenvalues of the stability matrix vanish only if the determinant of the matrix, $|f_{ij}| = 0$, the spinodal instability surface for our ternary system is given by

$$3\chi^2(c_{AC}c_{BC}) - 2\chi(c_{AC}c_B + c_{BC}c_C + c_{AC}c_C) + 1 = 0 \quad (4.1)$$

A section (cut) of this surface at $\chi = 3.5$ is schematically shown in Fig. 4.2, in which we delineate three distinct regions marked as *I*, *II* and *III*.

In region *I*, also referred to as the global spinodal region, both the eigenvalues of the matrix are negative and the solid solution is unstable with respect to local composition fluctuation along any direction as shown in Fig. 4.2. Note that P lies in this region. In contrast to region *I*, region *II* is characterized by only one negative eigenvalue (the other eigenvalue is positive). The alloys lying within this region are unstable only along certain directions in the composition space; alloys Q, R, S and T lie in this conditionally stable region. The directions for which a small amplitude perturbation leads to the steepest fall in the free energy is given by the eigenvector associated with the negative eigenvalue; these directions are shown in Fig. 4.2. Finally, in the region marked as *III*, the alloy is unconditionally stable with respect to a composition fluctuation in any direction.

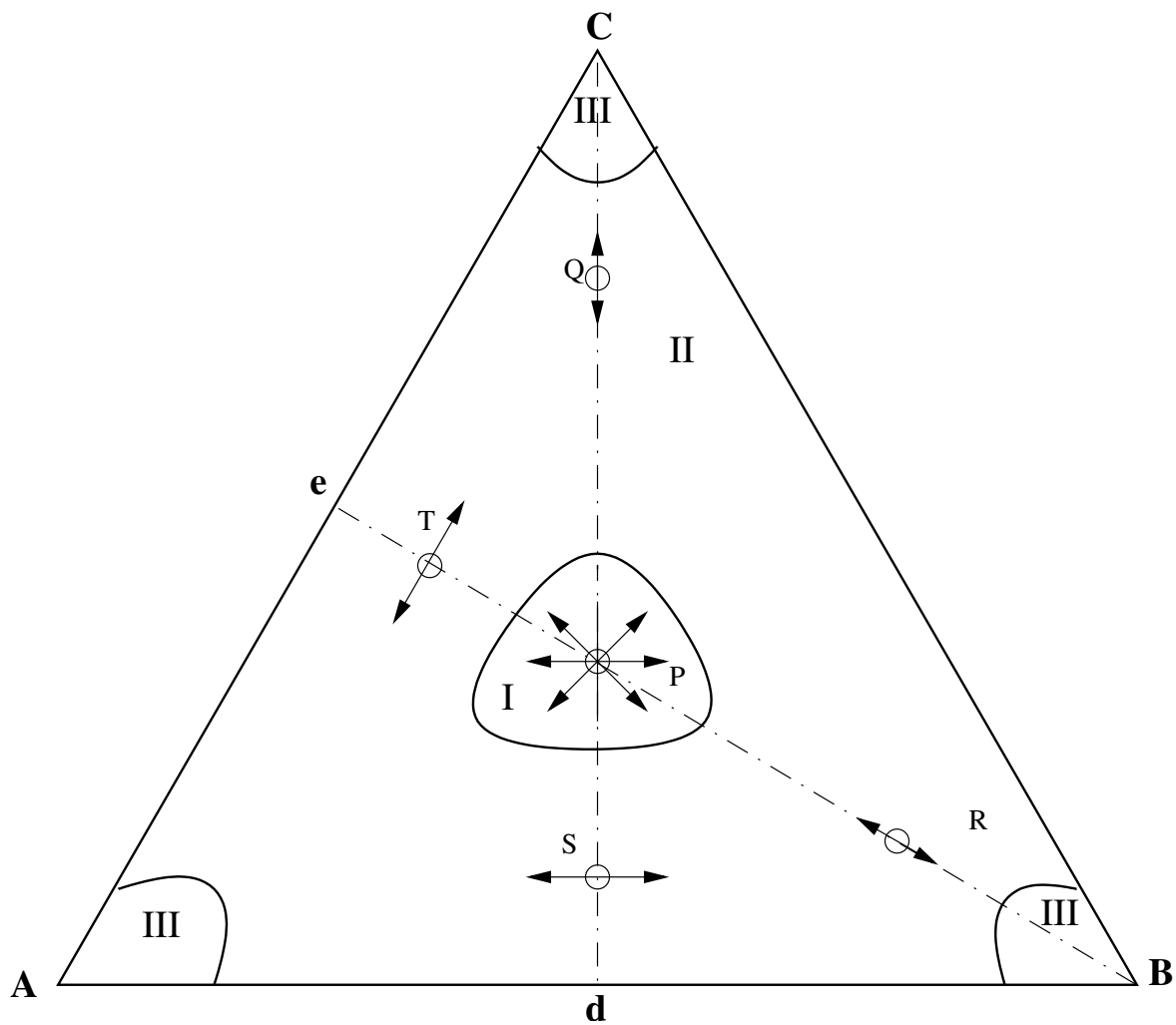


Figure 4.2: Isothermal section of the chemical spinodal surface for our ternary regular solution model with $\chi_{AB} = \chi_{BC} = \chi_{AC} = \chi = 3.5$ (schematic).

4.2 Choice of gradient energy parameters

One of the main objectives of our work is to study the effect of relative interfacial energies between the coexisting phases on the morphological evolution. The interfacial energy between any two of the coexisting phases may be varied systematically by changing the gradient energy coefficients, K_A , K_B and K_C , with the same bulk free energy of the system, given by $f(c_A, c_B, c_C)$ in Eqn. (3.2). For our simulations, we have chosen two systems, which differ only in the values of K_A , K_B and K_C , listed in Table 4.2. The interfacial

Table 4.2: **Gradient energy parameters and corresponding scaled interfacial energies**

System	K_A	K_B	K_C	$\sigma_{\alpha\beta}$	$\sigma_{\alpha\gamma}$	$\sigma_{\beta\gamma}$
1	4.0	4.0	4.0	1.195	1.195	1.195
2	4.0	32.0	4.0	2.628	1.236	2.628

energies, $\sigma_{\alpha\beta}$, $\sigma_{\beta\gamma}$ and $\sigma_{\alpha\gamma}$ in these two systems may be calculated as follows: For $\sigma_{\alpha\beta}$, for example, a one-dimensional simulation is set up with one half of the system with α and the other half with β , as the initial configuration; then the Cahn-Hilliard equations, Eqns. (3.27 and 3.28), are solved to the steady state. Such steady-state interfacial composition profiles are shown in Fig. 4.3, and their interfacial energies (in scaled form obtained using the procedure described below) are listed in Table 4.2.

The expression for interfacial energy between any two coexisting isotropic phases,

say, α and β of compositions c_i^α and c_i^β , assuming a flat interface, $\sigma_{\alpha\beta}$ is [2, 16]:

$$\sigma_{\alpha\beta} = N_v \int_{-\infty}^{\infty} \left(f^m + \sum_{i=A,B,C} K_{ii} \left(\frac{\partial c_i}{\partial x} \right)^2 \right), \quad (4.2)$$

where

$$f^m = f(c_i(x)) - \sum_i c_i(x) \mu_i^{\alpha/\beta}; \quad i = A, B, C. \quad (4.3)$$

$\mu_i^{\alpha/\beta}$ is the chemical potential of the i 'th species within the bulk of the α or the β phase which are in equilibrium with each other. In the scaled form, $\sigma_{\alpha\beta} = \sigma_{\alpha\beta}^* / N_v^* l^* e^*$. Similarly, $\sigma_{\alpha\gamma}$ and $\sigma_{\beta\gamma}$ are evaluated.

In system 1, $\alpha\beta$, $\beta\gamma$ and $\alpha\gamma$ interfaces are equivalent. Thus, the (ABC) profiles shown in Fig 4.3(a) for the $\alpha\beta$ interface are also valid for $\beta\gamma$ and $\alpha\gamma$ interface by replacing components (ABC) with (BCA) and (CAB), respectively. On the other hand, in system 2, $\alpha\beta$ and $\beta\gamma$ interfaces are equivalent, and different from the $\alpha\gamma$ interface. For the $\alpha\beta$ interface, the composition profile of the minority component, C is asymmetric about $x=0$; the interface is also much broader (see Fig. 4.3(b)). On the other hand, when B (with a larger gradient energy contribution) is the minority component, the amount of B segregating to the low-energy $\alpha\gamma$ interface is much smaller in Fig. 4.3(c) compared to that of C in Fig. 4.3(b).

Since the $\alpha\beta$, $\beta\gamma$ and $\alpha\gamma$ are equivalent in system 1 (see Table 4.2), the pair of alloys Q and R (see Fig. 4.1) chosen along the lines Cd and Be , respectively, are equivalent. Similarly, the pairs of alloys S and T are also equivalent in system 1. However, in system 2, the pairs of alloys (Q, R) and (S, T) are not equivalent, since the species B is distinct from the other two species (e.g., $K_B \neq K_A = K_C$); therefore, they are studied separately.

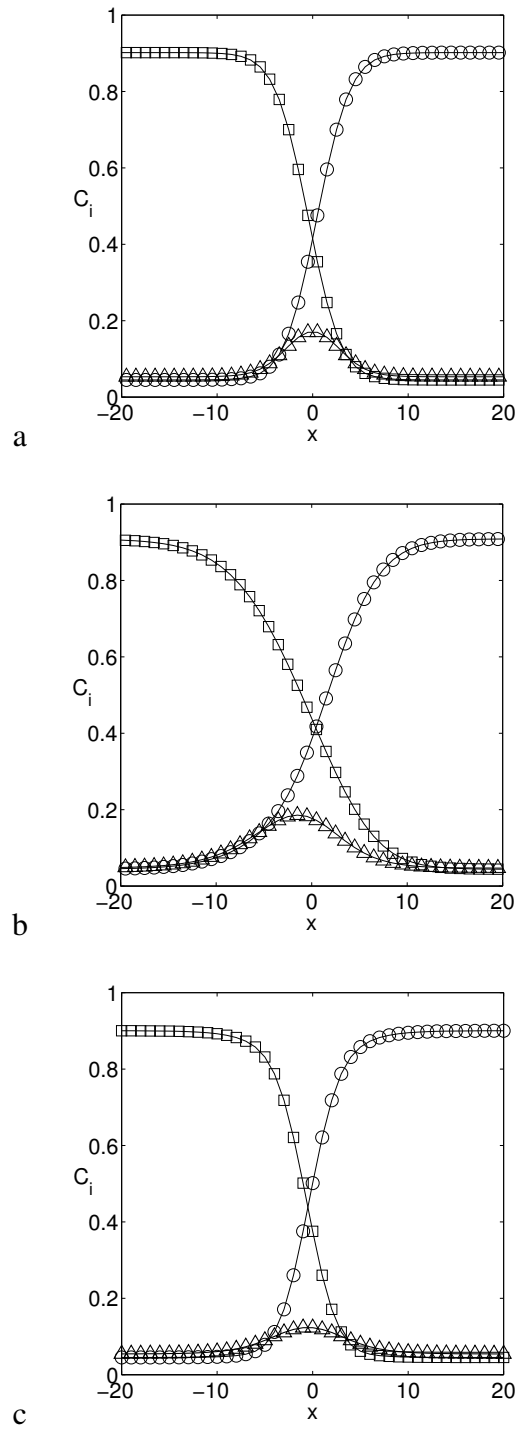


Figure 4.3: Equilibrium composition profile across (a) an $\alpha\beta$ interface in system 1 (with $\sigma_{\alpha\beta} = \sigma_{\beta\gamma} = \sigma_{\alpha\gamma}$), (b) an $\alpha\beta$ interface in system 2 (with $\sigma_{\alpha\beta} = \sigma_{\beta\gamma} \approx 2\sigma_{\alpha\gamma}$), (c) an $\alpha\gamma$ interface in system 2.

4.3 Simulation details

We carry out our simulations for the five different alloy compositions and the two sets of interfacial energies on a 256×256 square grid with a dimensionless grid spacing of $\Delta x = 1.0$ using periodic boundary conditions. Each simulation starts with a prescribed homogeneous composition, on which an initial uniform random fluctuation between -0.005 and $+0.005$ is introduced at each grid point. A dimensionless time step of $\Delta t = 0.5$ is used for integration. We have used the fast Fourier transform package ‘fftw’ developed by Frigo and Johnson [41]. In all simulations we have used scaled mobilities: $M_{AA} = 1.0$, $M_{BB} = 1.0$ and $M_{AB} = M_{BA} = -0.5$; these values ensure that the mobility matrix is positive definite (see also section 3.3).

The results of microstructural evolution are presented as time snapshots using a gray scale representation of the local concentrations of the species A , B and C . A gray scale map of the Gibbs triangle is shown in Fig. 4.4. The Gibbs triangle is divided into four equal area triangles and each subtriangle is represented by a different gray shade. The lower left triangle has the lightest shade, the lower right triangle has a dark shade, the top triangle is darker and the central triangle is the darkest. The gray scale suitably represents the concentrations of A , B and C within the bulk of the phases as well as at the interface between the phases.

The following convention has been used for denoting the alloys: the uppercase letter represents the composition of the alloy in Fig. 4.1 and the subscript is used to distinguish between the two different systems ‘1’ and ‘2’. For example, the composition of alloy P_1 is represented by the point P in Fig. 4.1 and the subscript ‘1’ corresponds to system 1 in which the interfacial energies between the coexisting phases are equal.

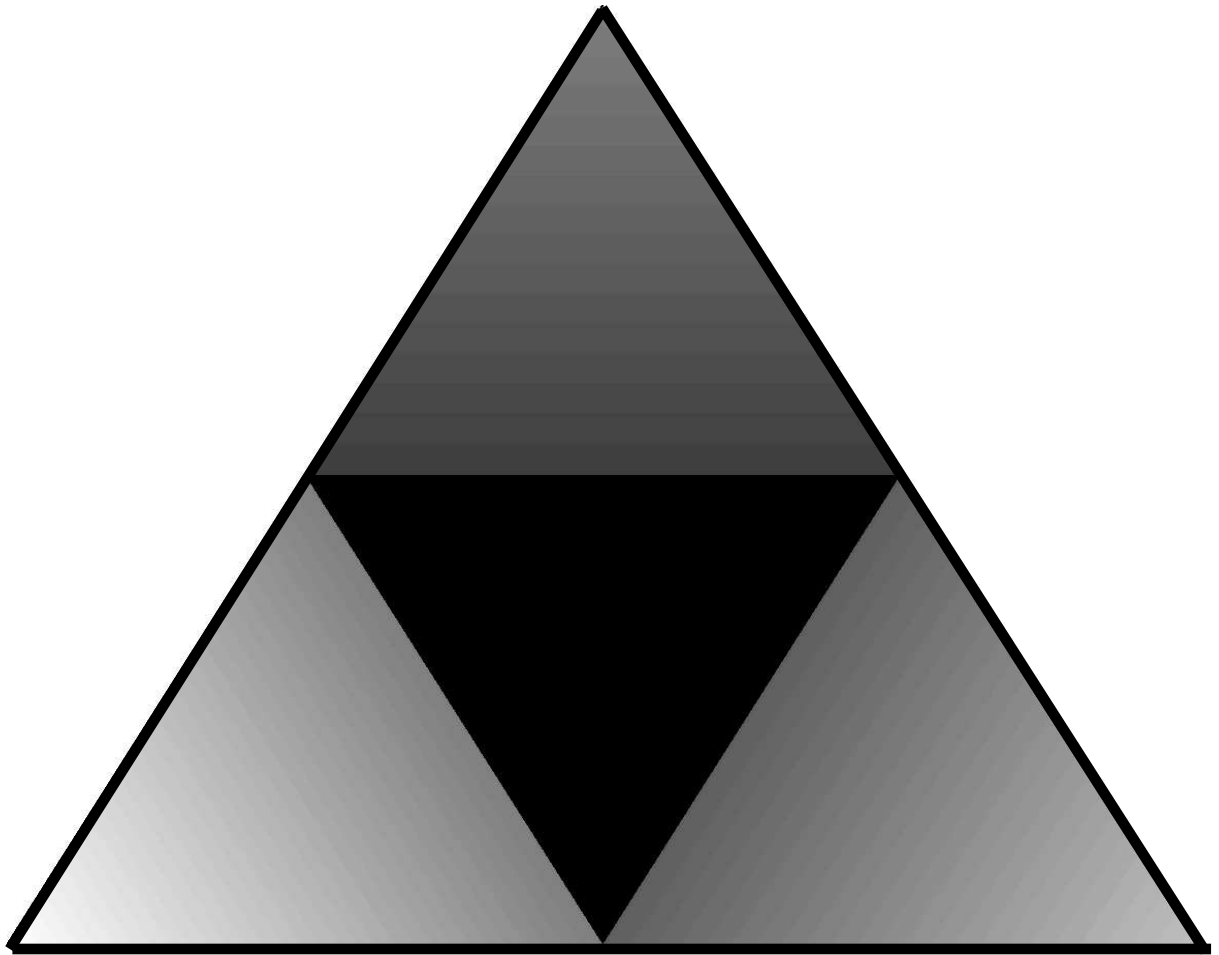


Figure 4.4: Gray scale color map projected on Gibbs triangle

4.4 Alloys P_1 and P_2

Microstructure

From the point of view of bulk thermodynamics alloys P_1 and P_2 do not differ in any way and lie inside the global spinodal region I (see Fig. 4.2). Therefore, composition fluctuation along *any direction* can grow. The microstructures of P_1 and P_2 at various times are shown in Figs. 4.5 and 4.6 respectively.

In alloy P_1 , at the initial stages of decomposition all the three phases, α (A-rich), β (B-rich) and γ (C-rich), start appearing simultaneously in the form of discrete particles, as shown in the microstructure at time $t = 400$. At $t = 500$ the process of decomposition has progressed to a considerable extent. At later times ($t = 1000, 2000$), the size of the particles become larger due to multiphase coarsening. Since the composition is completely symmetric with respect to the three components, the three phases, α , β and γ have similar morphologies.

In alloy P_2 , since the β phase is associated with a much larger interfacial energy, it would be expected to appear later in this alloy than in P_1 . It does indeed appear later, as can be seen from the microstructures shown in Fig. 4.6. At early times, $t = 400$, the microstructure appears black because of the extremely sluggish nature of decomposition. At $t = 500$, only α and γ phases start appearing in the form of discrete particles while the formation of β phase is suppressed until this time. Further, the α and γ particles that appear are distinctively fewer in number as compared to alloy P_1 , and occupy only a smaller fraction of the system than in P_1 . At $t = 1000$, the microstructure consists of nearly continuous chains of alternating α and γ particles surrounding the β particles.

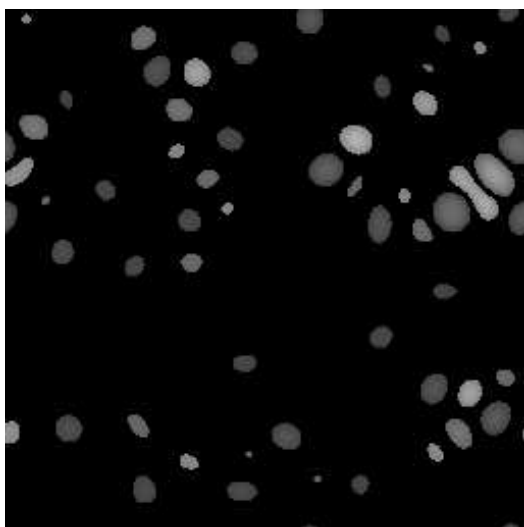
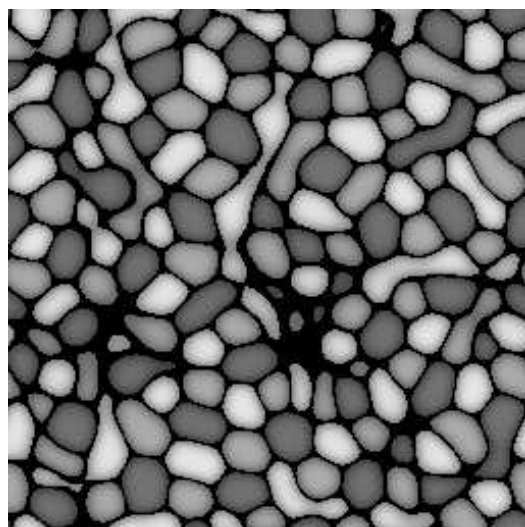
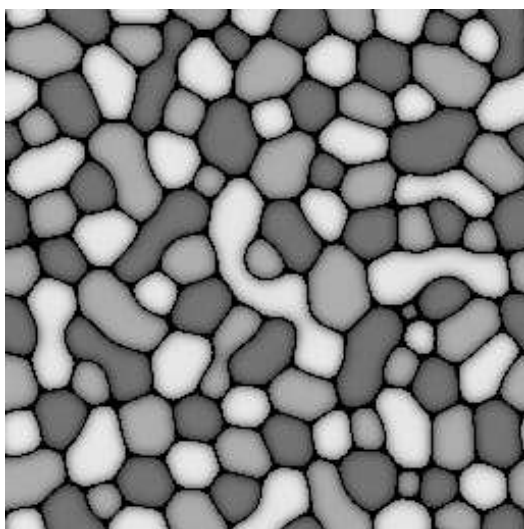
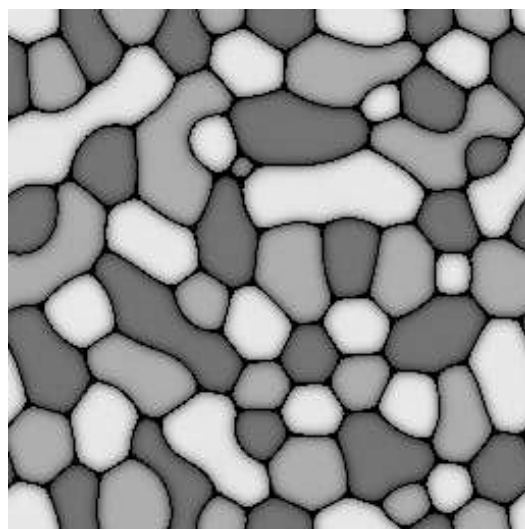
 $t=400$  $t=500$  $t=1000$  $t=2000$

Figure 4.5: Microstructural evolution in alloy P_1 .

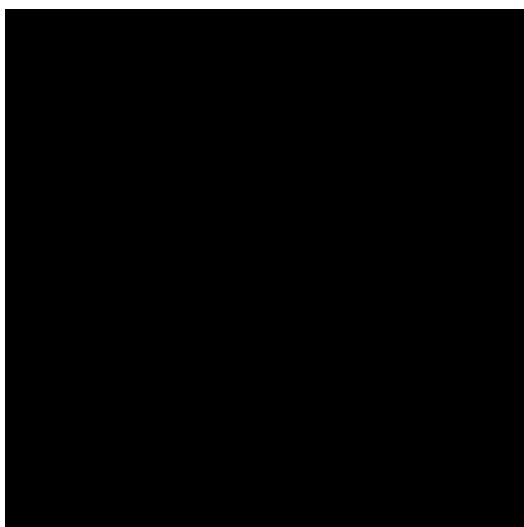
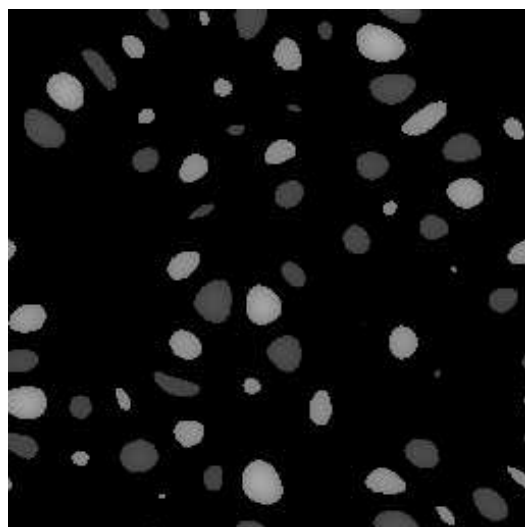
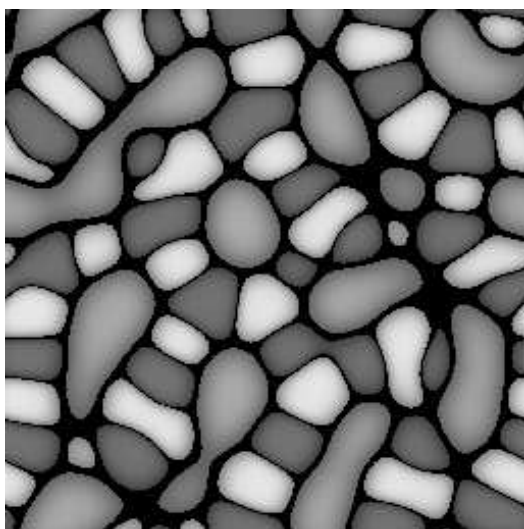
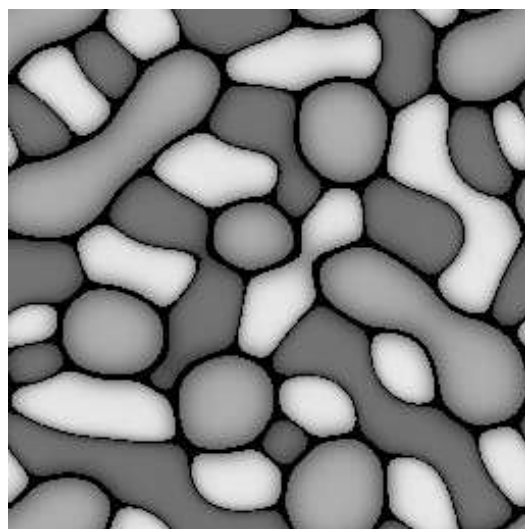
 $t=400$  $t=500$  $t=1000$  $t=2000$

Figure 4.6: Microstructural evolution in alloy P_2 .

Comparing the length scale of microstructures at $t = 1000$ and $t = 2000$ with those in Fig. 4.5 for alloy P_1 , we see that the individual A-rich, B-rich and C-rich domains are much coarser in P_2 , due to the higher interfacial energy associated with these particles.

Kinetic paths of α , β and γ phases

In order to follow the course of spinodal decomposition in alloys P_1 and P_2 , we plot the kinetic paths (i.e., compositional history) of the α , β and γ phases and project these paths onto the Gibbs triangle (see Figs. 4.7 and 4.8). For this purpose, we track the composition of some representative A-rich, B-rich and C-rich regions in the microstructure from the beginning of the simulation till the time at which the decomposition is nearly complete ($t \sim 1000$).

For P_1 , we observe that the kinetic paths of all the three phases are similar and are in good agreement with the predictions based on de Fontaine's stability analysis [34]. For P_2 , the kinetic paths of α and γ phases are significantly different from that of β phase, as shown in Fig. 4.7. It is clear that partitioning of species B is absent during early stages due to the higher interfacial energy associated with the B-rich β phase. Thus, initial decomposition primarily takes place along a line parallel to the AC direction producing A-rich and C-rich regions in the microstructure. Coarse B-rich regions form only later during further decomposition. This clearly indicates that the history of A-rich, B-rich and the C-rich regions is dependent not only on the starting alloy composition but also on the relative interfacial energies of the three phases.

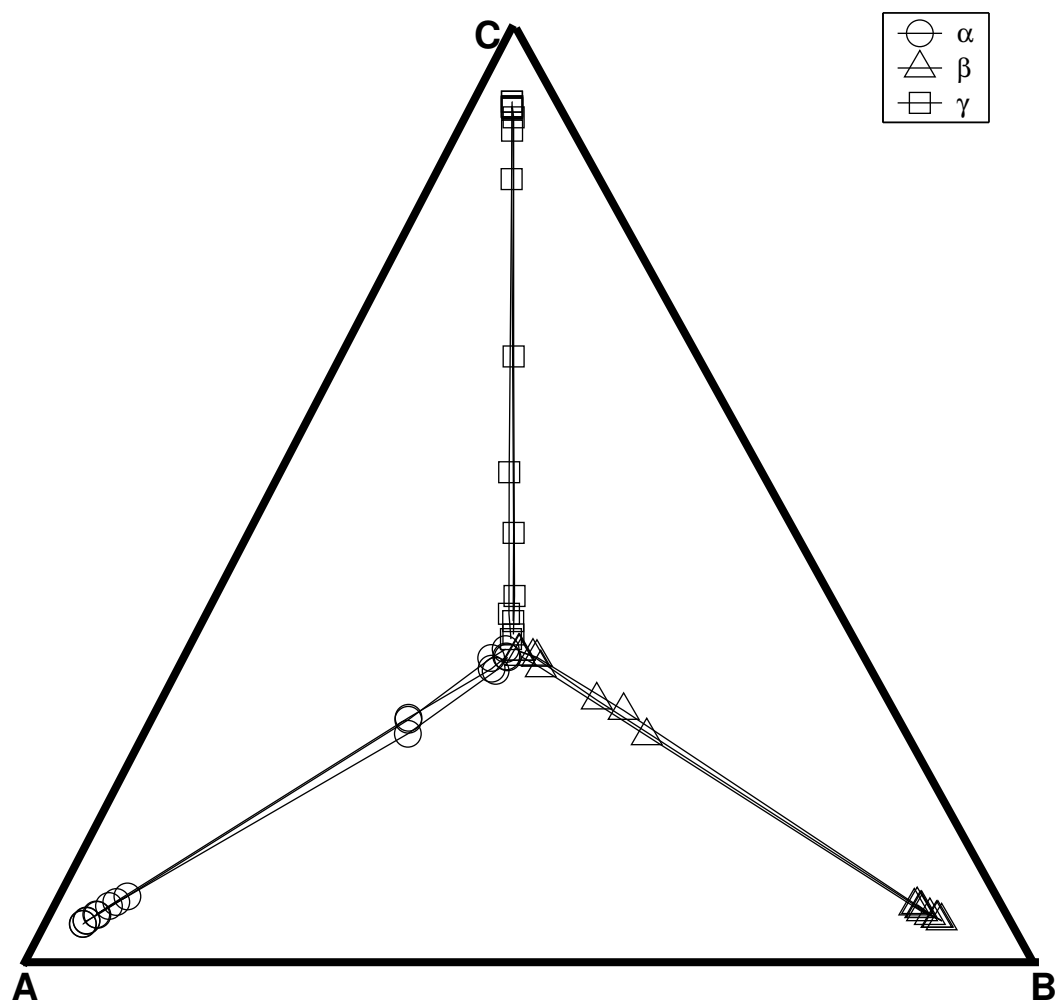


Figure 4.7: Kinetic paths of α , β and γ phases in alloy P_1 ($\sigma_{\alpha\beta} = \sigma_{\beta\gamma} = \sigma_{\alpha\gamma}$); (lines connecting the points is only a guide to the eye.)

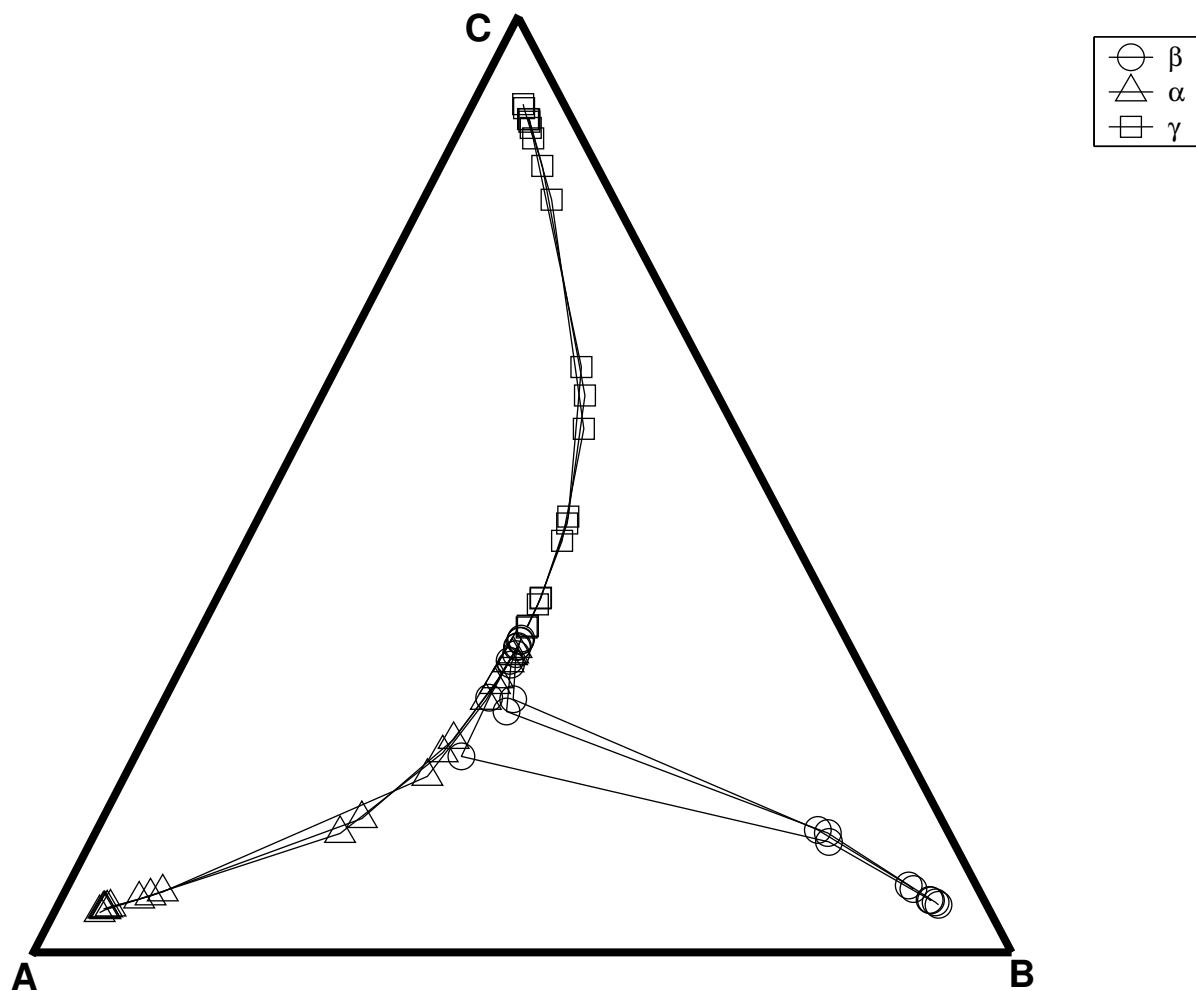


Figure 4.8: Kinetic paths of α , β and γ phases in alloy P_2 ($\sigma_{\alpha\beta} = \sigma_{\beta\gamma} \approx 2\sigma_{\alpha\gamma}$); (lines connecting the points is only a guide to the eye.)

Structure functions

Apart from the kinetic paths, the three phase morphologies for alloys P_1 and P_2 are characterized by three linearly independent time-dependent structure functions $S_{ii}(\mathbf{k}, t)$, defined by:

$$S_{ii}(\mathbf{k}, t) = \frac{1}{N} \left\langle \sum_r \sum_{r'} e^{-i\mathbf{k} \cdot \mathbf{r}} [c_i(\mathbf{r} + \mathbf{r}', t) - \bar{c}_i] [c_i(\mathbf{r}', t) - \bar{c}_i] \right\rangle; \quad i = A, B, C,$$

where $\bar{c}_i$ is the average concentration of species i in the alloy. Both the sums run over the entire lattice. $k = |\mathbf{k}|$ is the magnitude of the scattering vector and is equal to the reciprocal of the wavelength of the composition modulation. $S_{ii}(\mathbf{k}, t)$ measures the strength of the spatial correlations in composition of component i with wave vector $\mathbf{k}$ and time t . The scattered intensity obtained from small angle (X-ray and neutron) scattering experiments (SAXS, SANS) for a ternary solid solution is expressed as a linear combination of these three structure functions. Averages are performed over three simulation runs, each run starting with a different set of random numbers for each initial state (i.e., a different initial composition field with the same average composition).

Since the system is isotropic, $S_{ii}(\mathbf{k}, t)$ depends only on the radial distance, $k_r = |\mathbf{k}|$, in reciprocal space. Hence, we plot the circularly averaged structure functions $S(k, t)$ for each of the alloys considered.

The structure functions exhibit some common features for all the alloys: S_{AA} , S_{BB} and S_{CC} curves are broad and shallow at earlier times; with increasing time, they become sharper and narrower. Further, as time increases, the peak value of the structure functions increase and shift to a lower value of k , indicating an increase in the average ‘real space’ length scale of the A-rich, B-rich and the C-rich domains. At later stages, the shift is significant due to coarsening of the individual domains.

For alloy P_1 , S_{AA} , S_{BB} and S_{CC} behaviours are as described above, and are essentially similar to each other. There are some minor differences, especially at later times, due largely to a small system size used in our simulations, and averaging over only 3 independent runs.

For alloy P_2 , S_{AA} and S_{CC} are more or less similar to those in P_1 ; the peak positions are shifted a little towards left in P_2 indicating the presence of coarser A-rich and C-rich domains. The major difference in the structure functions for the two systems is observed in S_{BB} . For P_2 , formation of B-rich clusters is delayed till $t = 1000$; e.g., at $t = 300, 400, 500$, S_{BB} curves for this alloy nearly graze the x-axis; this observation supports our earlier conclusion (see Figs. 4.9 and 4.10) that partitioning of B is virtually absent during the early stages.

The peak in S_{BB} , once it starts appearing at $t = 1000$, increases with time more rapidly than for S_{AA} or S_{CC} . This implies that coarsening of β particles is faster than those of α and γ particles. This conclusion will also be corroborated using kinetic data presented in section 4.7.

4.5 Alloys Q_1 , Q_2 and R_2

Microstructures

In alloy Q_1 , the composition modulations are along the line Cd in Fig. 4.2, because the driving force for phase separation along the Cd direction is much higher than along any other direction. So, at short times ($t = 300$) microstructure exhibits a phase separation process (see Fig. 4.11) which produces an interconnected two-phase microstructure con-

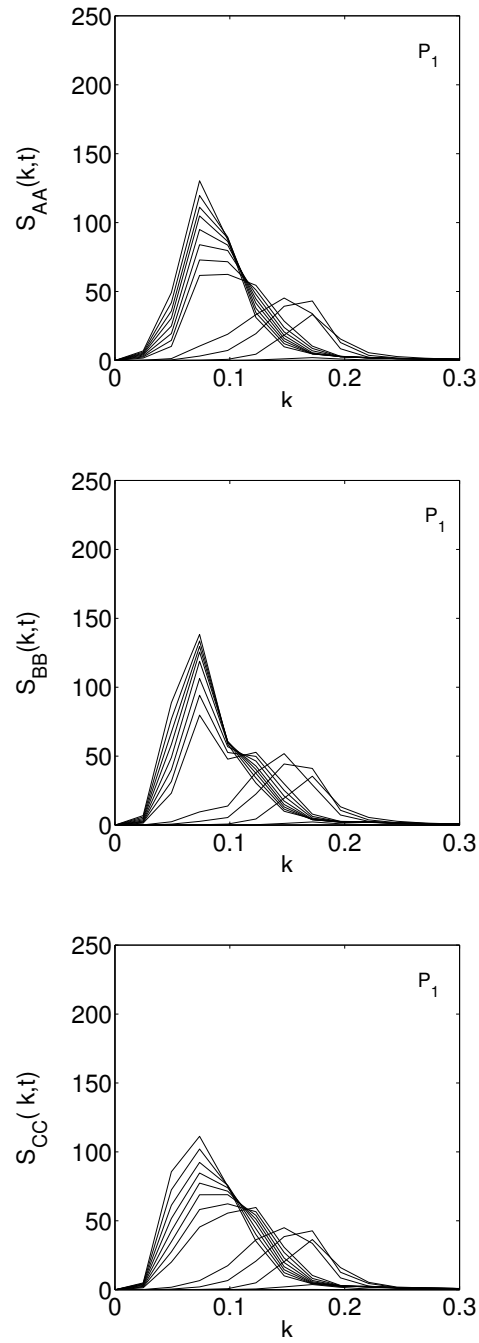


Figure 4.9: Structure functions S_{AA} , S_{BB} , S_{CC} plotted as a function of k for alloy P_1 . In each figure, the times are for $t=300, 400, 500, 1000, 2000, 2250, 2500, 2750, 3000, 3250, 3500, 3750$.

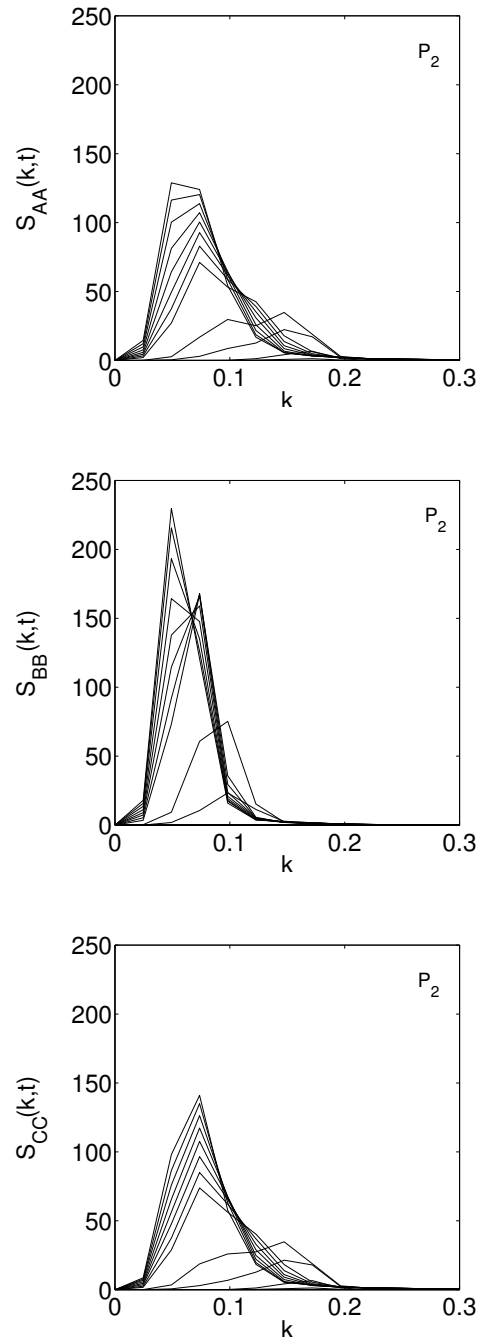


Figure 4.10: Structure functions S_{AA} , S_{BB} , S_{CC} plotted as a function of k for alloy P_2 . In each figure, the times are for $t=300, 400, 500, 1000, 2000, 2250, 2500, 2750, 3000, 3250, 3500, 3750$.

sisting of stable C-rich γ phase and an unstable C-poor (or AB-rich) phase. At a later time, ‘AB-rich’ phase decomposes further into A-rich and B-rich phases resulting in the formation of a chain of alternating beads of α and β particles distributed inside a continuous γ matrix. The chains also exhibit extensive branching, which remains up to $t = 3000$.

In alloy Q_2 , the initial decomposition results in the formation of a C-rich γ phase and a C-poor phase (see Fig. 4.12). But the phases are not interconnected and the γ phase has a much higher volume fraction than the C-poor phase ($t = 300$). However, the C-rich and C-poor phases have nearly the same B-content (this observation will be substantiated later using structure function results). The onset of formation of A-rich α phase is observed at $t = 500$ while formation of B-rich β phase is delayed to $t > 500$ due to the higher interfacial energy associated with it. Once again, a beaded microstructure is observed with the α and the β phases forming the beads. However, their morphology is different from that in alloy Q_1 ; beads of β phase are bigger and more rounded than those of the α phase.

In alloy R_2 (rich in B), the composition modulations occur along the Be direction (see Fig. 4.2). So, initial decomposition leads to the formation of stable B-rich β phase and an unstable B-poor phase at $t = 300$ (Fig. 4.13). If we compare the length scales in this figure and in Fig. 4.11 for Q_1 , we observe a coarser microstructure for R_2 at all times. The domains remain interconnected before the B-poor phase begins (after $t = 1000$) to decompose into A-rich and C-rich phases. At $t = 3000$, the microstructure contains alternating beads of α and γ phases in a continuous β matrix. Their morphology is different from that in alloy Q_1 : the chains show very little branching in alloy R_2 for $t \geq 2000$.

Structure functions

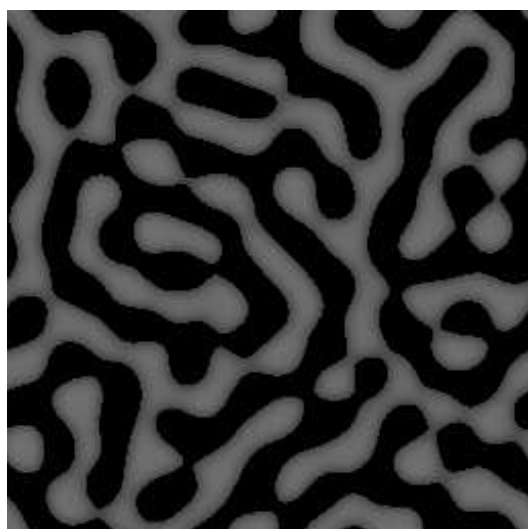
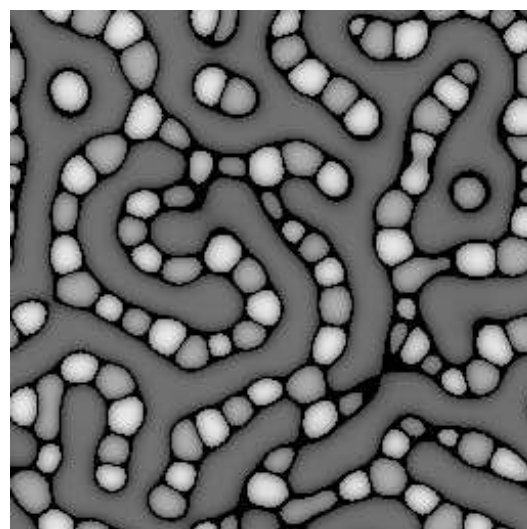
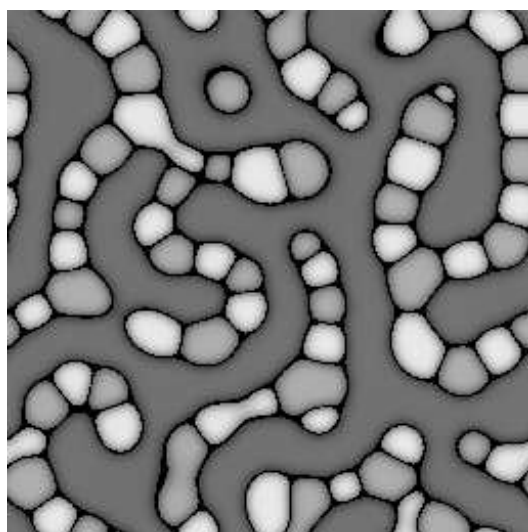
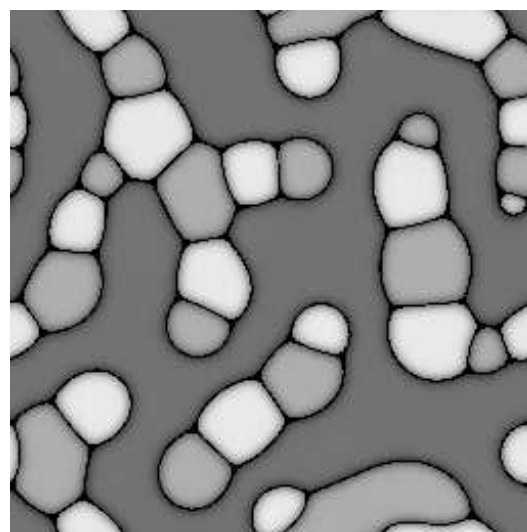
 $t=300$  $t=500$  $t=1000$  $t=3000$

Figure 4.11: Microstructural evolution in alloy Q_1 .

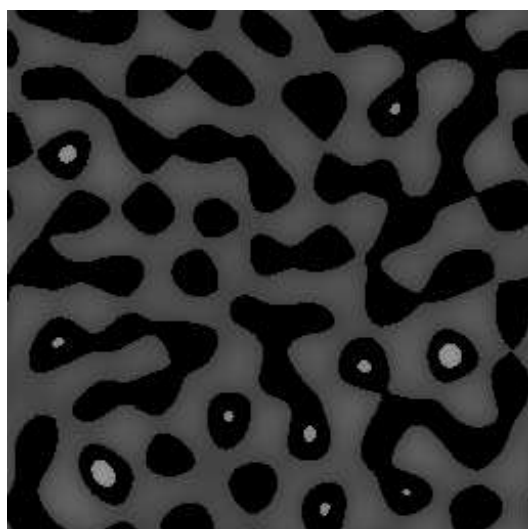
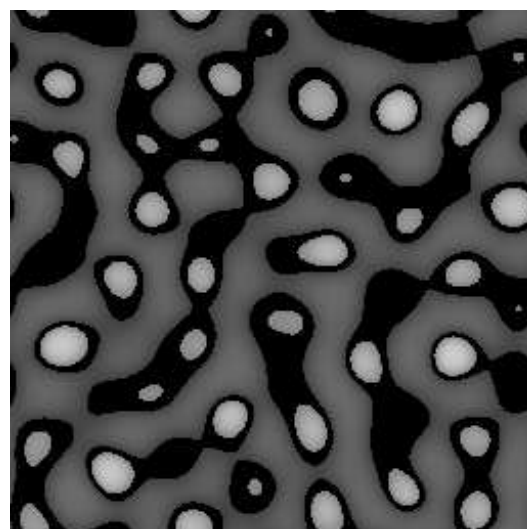
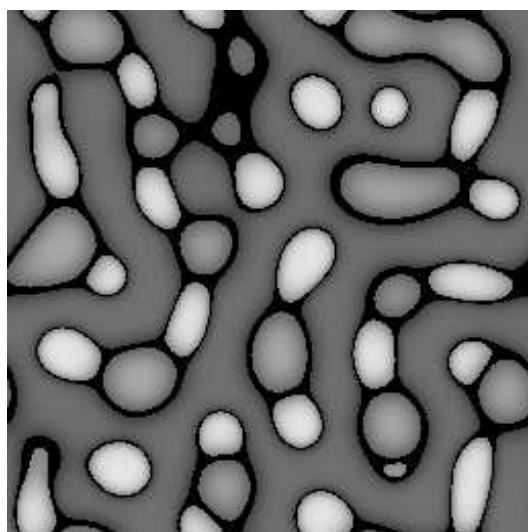
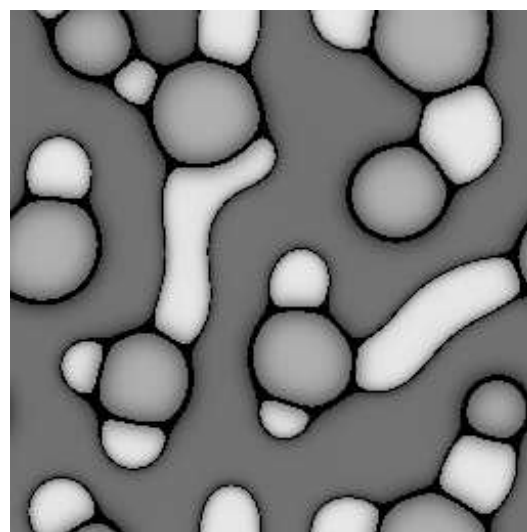
 $t=300$  $t=500$  $t=1000$  $t=3000$

Figure 4.12: Microstructural evolution in alloy Q_2 .

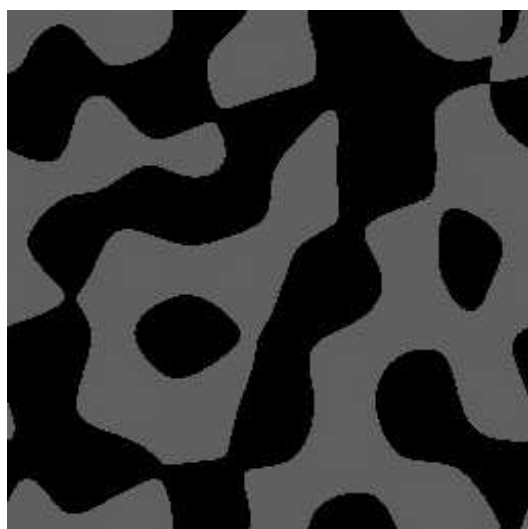
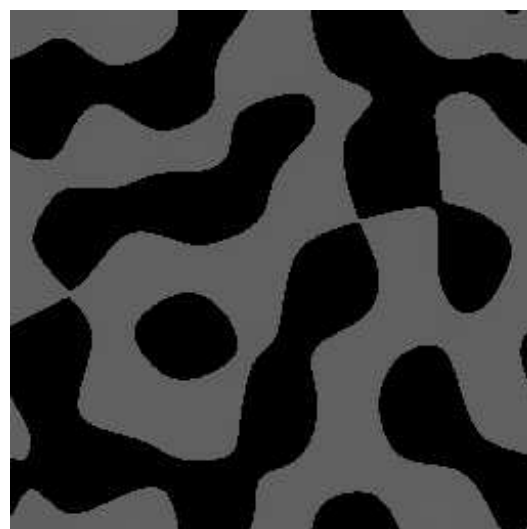
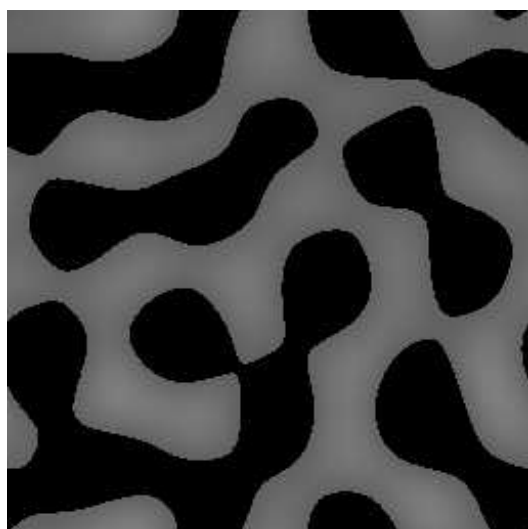
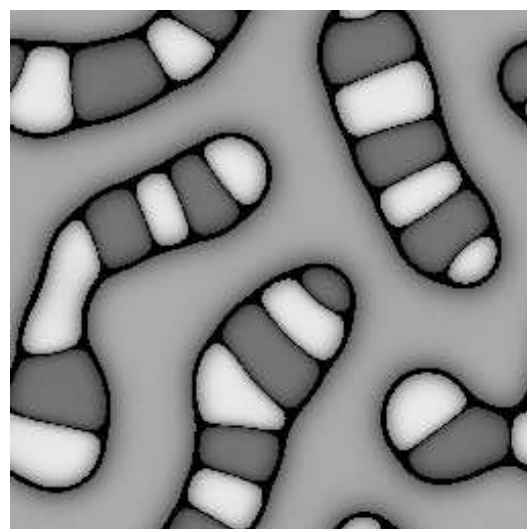
 $t=300$  $t=500$  $t=1000$  $t=3000$

Figure 4.13: Microstructural evolution for alloy R_2

The circularly averaged structure functions for alloys Q_1 , Q_2 and R_2 are shown in Figs. 4.14, 4.15 and 4.16, respectively. For alloy Q_1 , S_{AA} and S_{BB} are similar. The weighted area occupied by S_{CC} is much bigger than S_{AA} and S_{BB} due to its high C-content. In contrast to alloy Q_1 , S_{AA} and S_{BB} for alloy Q_2 are different. Initial decomposition in alloy Q_2 is pseudo binary, with no partitioning of species B until $t = 1000$. Therefore, at times $t = 300, 400, 500$, S_{BB} for Q_2 coincides with the x-axis. Moreover, the B-rich domains are coarser in Q_2 than in Q_1 as indicated by the peak positions corresponding to S_{BB} in this alloy.

The species B and C in alloy R_2 are equivalent to species C and B, respectively, in alloy Q_2 . Since, partitioning of species B takes place right from the early stages, the decomposition is quite sluggish in alloy R_2 ; in Fig. 4.16, S_{AA} , S_{BB} and S_{CC} curves merge with the x-axis for $t \leq 1000$. The B-rich domains appear at later times in this alloy and are coarser than the corresponding C-rich domains in Q_1 and Q_2 , as indicated by the peak positions of S_{BB} at later times.

4.6 Alloys S_1 , S_2 and T_2

Microstructure

In alloy S_1 , whose microstructural evolution is shown in Fig. 4.17, the composition modulations develop along the line AB , as indicated by the arrowheads in Fig. 4.2. The initial decomposition ($t = 300$) leads to an interconnected two-phase microstructure consisting of A-rich α and B-rich β phases while the minority component C segregates to the interphase boundaries between these two phases. The pseudo-binary phase separation is

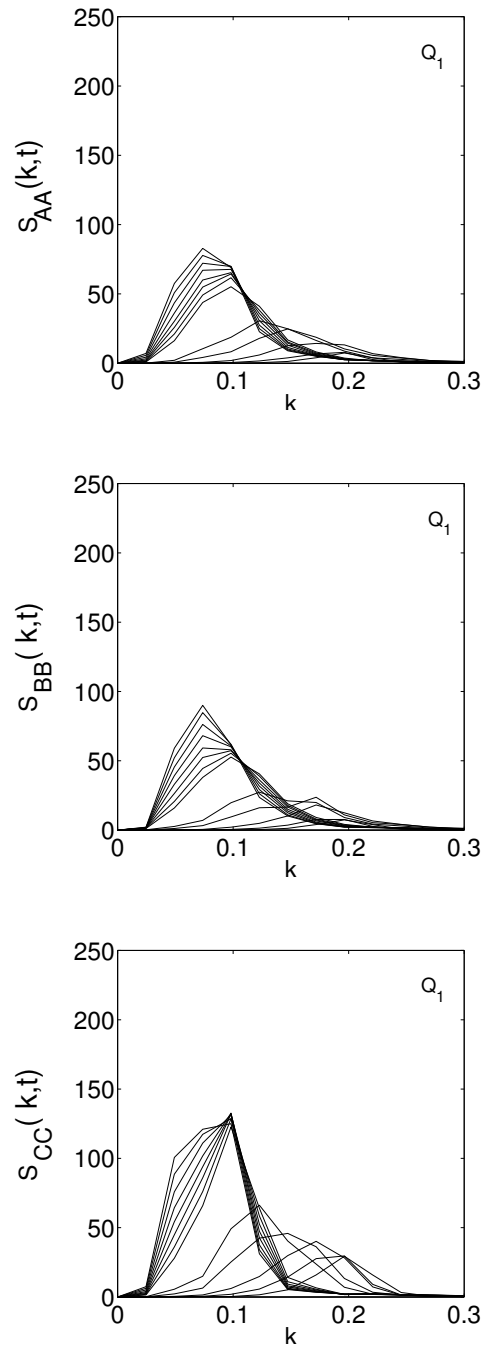


Figure 4.14: Structure functions S_{AA} , S_{BB} , S_{CC} plotted as a function of k for alloy Q_1 . In each figure, the times are for $t=300, 400, 500, 1000, 2000, 2250, 2500, 2750, 3000, 3250, 3500, 3750$

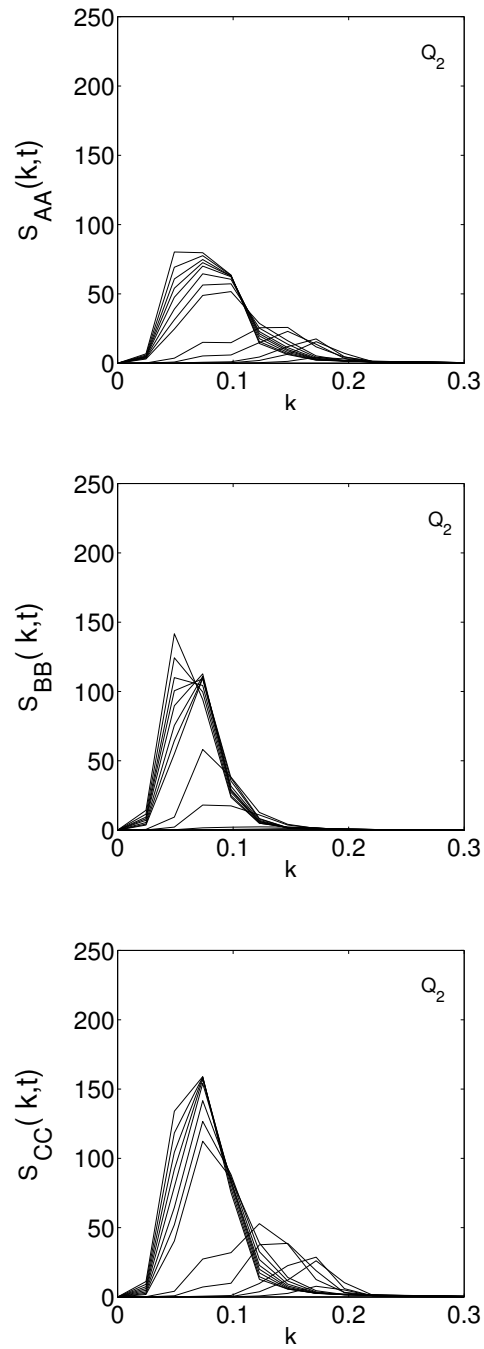


Figure 4.15: Structure functions S_{AA} , S_{BB} , S_{CC} plotted as a function of k for alloy Q_2 . In each figure, the times are for $t=300, 400, 500, 1000, 2000, 2250, 2500, 2750, 3000, 3250, 3500, 3750$.

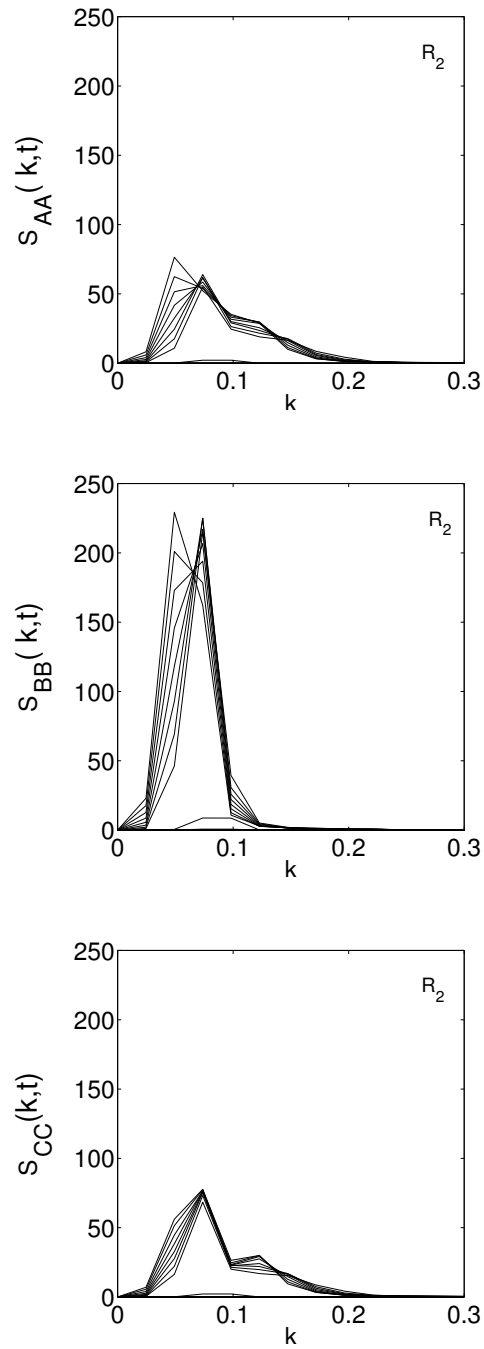


Figure 4.16: Structure functions S_{AA} , S_{BB} , S_{CC} plotted as a function of k for alloy R_2 . In each figure, the times are for $t=300, 400, 500, 1000, 2000, 2250, 2500, 2750, 3000, 3250, 3500, 3750$.

followed by coarsening of the interconnected domains ($t = 500$) and secondary phase separation; as a result of the latter, small inclusions of the third phase (richer in minority component C) start appearing at the $\alpha\beta$ interfaces. At later stages of decomposition ($t = 3000$), the morphology consists of almost continuous α and β phases with the γ phase particles occupying the interfacial regions.

In alloy S_2 , the initial decomposition into A-rich and B-rich regions is slow since it involves formation of B-rich regions with a higher interfacial energy. This process also leads to coarser features in the microstructure than in alloy S_1 (compare Figs. 4.17 and 4.18 for $t=300$ and $t=500$). Moreover, the B-rich regions have a much darker shade than that of the equilibrium β particles, indicating a larger C-content in them. The formation of C-rich regions through further decomposition is also correspondingly delayed to well after $t = 1000$; it leads to γ particles at the $\alpha\beta$ boundaries. At $t = 3000$, the microstructure consists of continuous tubes of α and β phases with fewer and coarser C-rich γ particles (compared to S_1) at the $\alpha\beta$ interphase boundaries.

The initial decomposition of the alloy T_2 takes place primarily along the AC direction as shown in Fig. 4.2. B-partitioning is absent at the initial stages of decomposition. The pseudo-binary phase separation at $t = 300$ produces an interconnected microstructure (see Fig. 4.19) containing A-rich and C-rich phases. At the later stages, the two-phase microstructure undergoes coarsening until well after $t = 1000$, when further decomposition leads to the formation of coarse β particles at the $\alpha\gamma$ interface. Though the length scale of α and γ phase regions of the microstructure at $t = 3000$ is only slightly larger than that in alloy S_1 , fewer and larger β particles are observed in S_2 than in S_1 even at the late stages of decomposition.

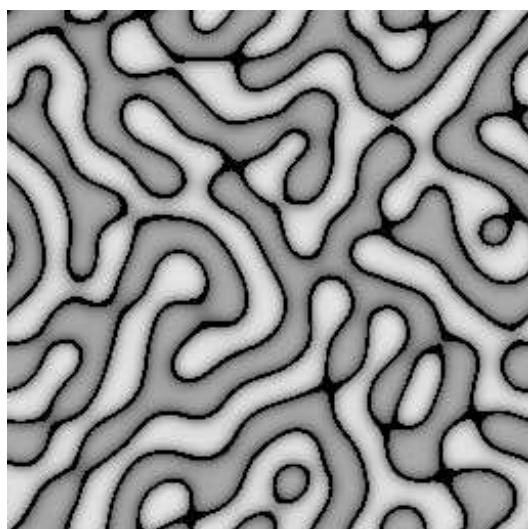
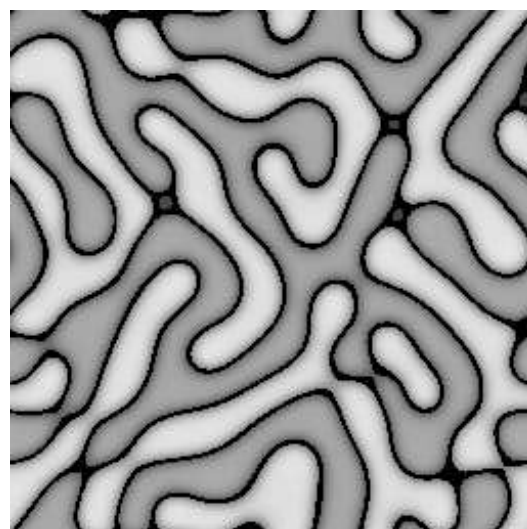
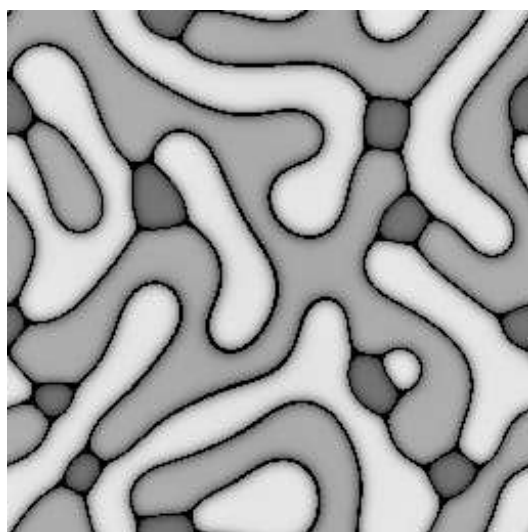
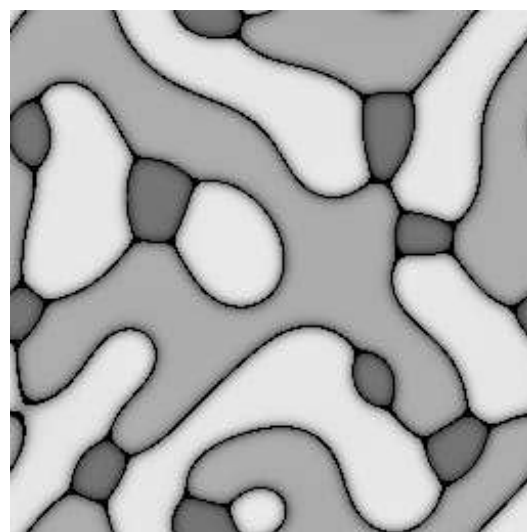
 $t=300$  $t=500$  $t=1000$  $t=3000$

Figure 4.17: Microstructural evolution for alloy S_1 .

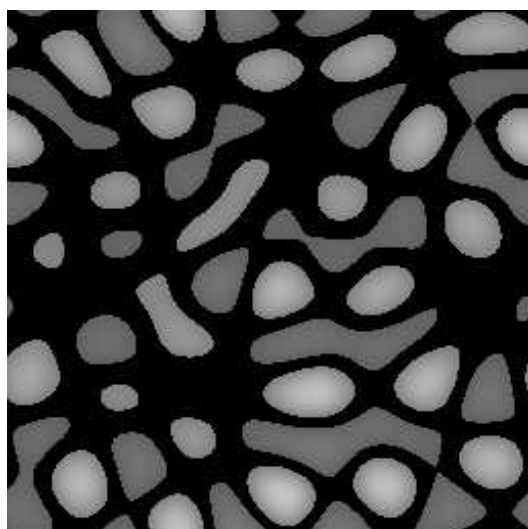
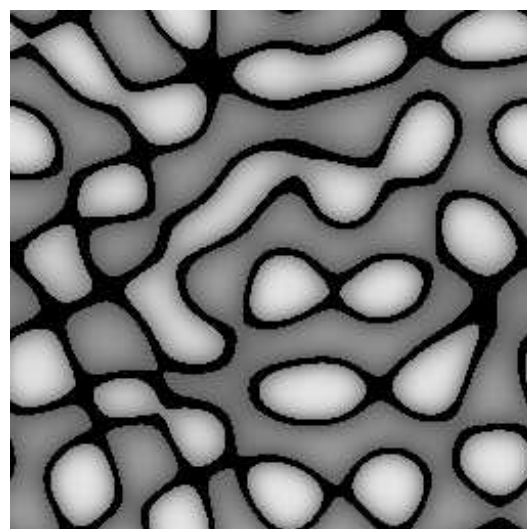
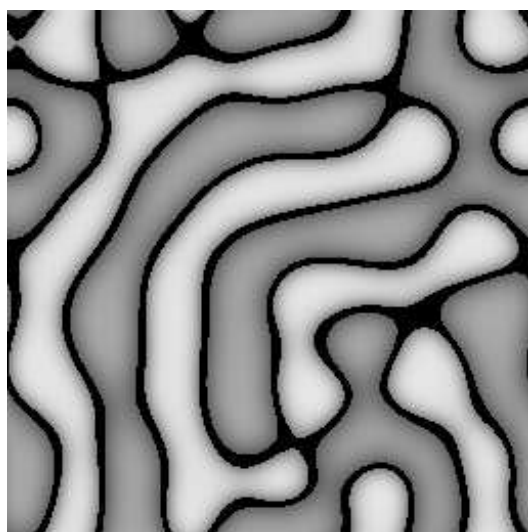
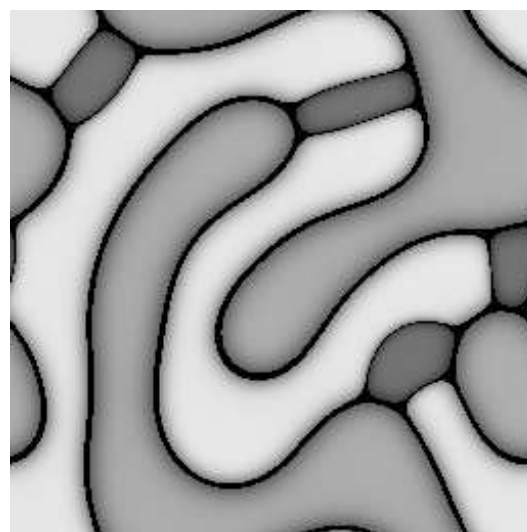
 $t=400$  $t=500$  $t=1000$  $t=3000$

Figure 4.18: Microstructural evolution for alloy S_2 .

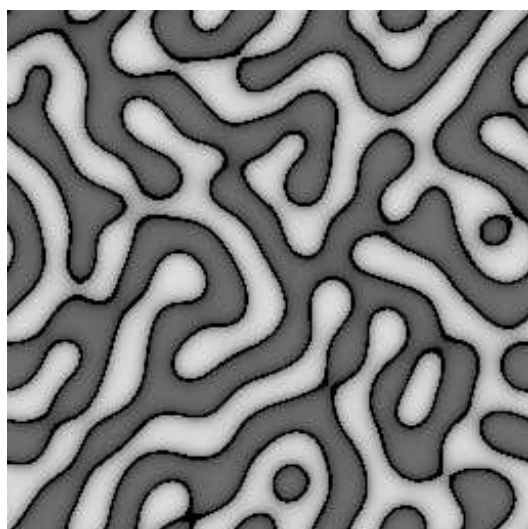
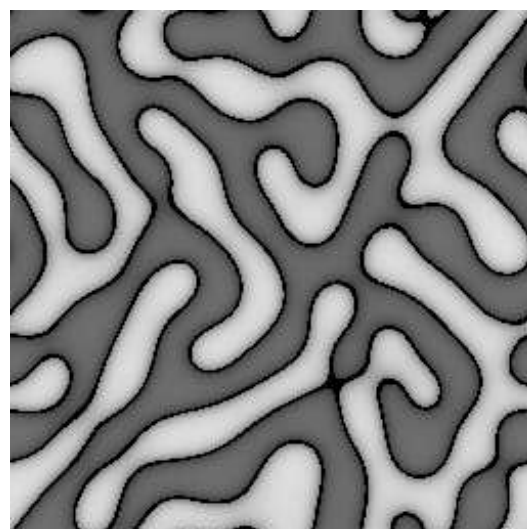
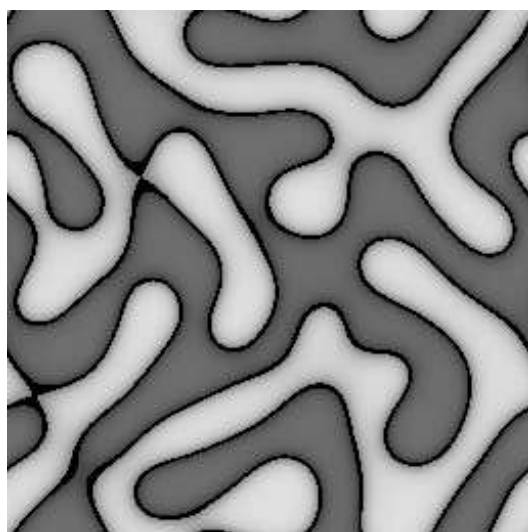
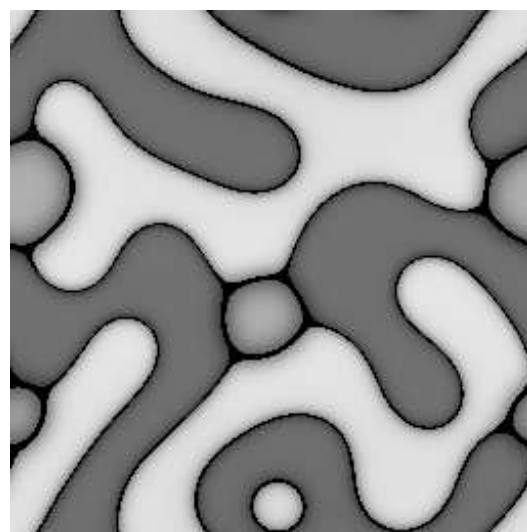
 $t=300$  $t=500$  $t=1000$  $t=3000$

Figure 4.19: Microstructural evolution for alloy T_2 .

Structure functions

The structure functions for alloys S_1 , S_2 and T_2 are shown in Figs. 4.20, 4.21, and 4.22. S_{AA} and S_{BB} for S_1 have nearly identical characteristics at all times. S_{CC} , corresponding to the minority component C in S_1 , coincides with the x-axis at times $t = 300, 400$ indicating a delay in the formation of C-rich domains in the microstructure until later times, for which S_{CC} shows a well-defined peak.

Careful comparison of the peak positions of the structure functions for S_2 with that of S_1 reveal that the A-rich, B-rich and the minority C-rich domains in S_2 have a coarser length scale. The formation of C-rich domains is delayed further until $t > 2000$ as can be seen from the absence of S_{CC} curves at earlier times. The degree of decomposition of the alloy with respect to C is also less in the alloy S_2 compared with that in alloy S_1 , as indicated by the flatness of the S_{CC} curves at all times.

For alloy T_2 (richer in C and poorer in B unlike S_1 and S_2), the structure functions, S_{AA} and S_{CC} (corresponding to the majority A-rich and C-rich domains) are similar to S_{AA} and S_{BB} for the alloy S_1 ; the formation of B-rich domains is considerably delayed in this alloy; the S_{BB} curves coincide with the x-axis at early times ($t = 300, 400, 500, 1000$).

4.7 Kinetics

During the late stages of decomposition, significant coarsening is observed in microstructures and structure functions. The characteristic length, R is a measure of the typical domain size and at late stages increases with time following a temporal power-law: $R^n \propto t$. The rate of growth of the individual domains depend on the actual coarsening mechanism.

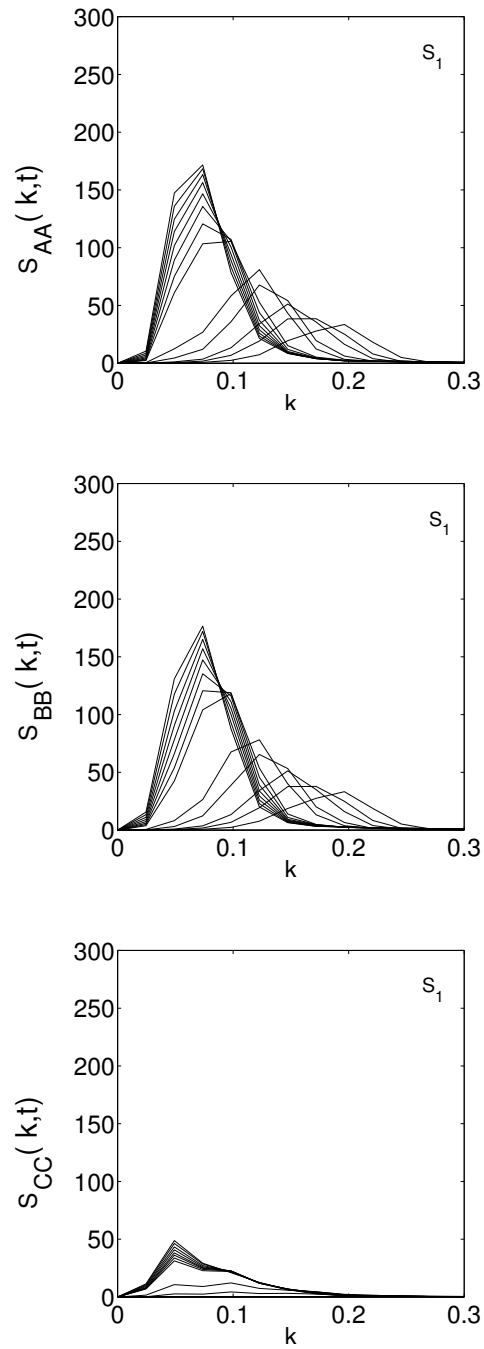


Figure 4.20: Structure functions S_{AA} , S_{BB} , S_{CC} plotted as a function of k for alloy S_1 . In each figure, the times are for $t=300, 400, 500, 1000, 2000, 2250, 2500, 2750, 3000, 3250, 3500, 3750$.

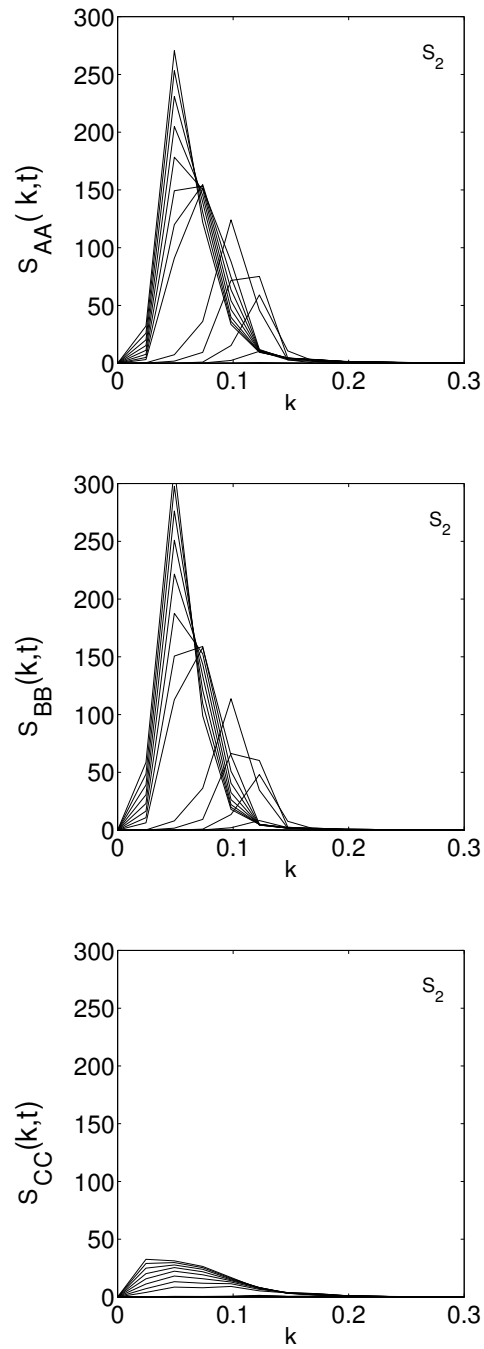


Figure 4.21: Structure functions S_{AA} , S_{BB} , S_{CC} plotted as a function of k for alloy S_2 . In each figure, the times are for $t=300, 400, 500, 1000, 2000, 2250, 2500, 2750, 3000, 3250, 3500, 3750$.

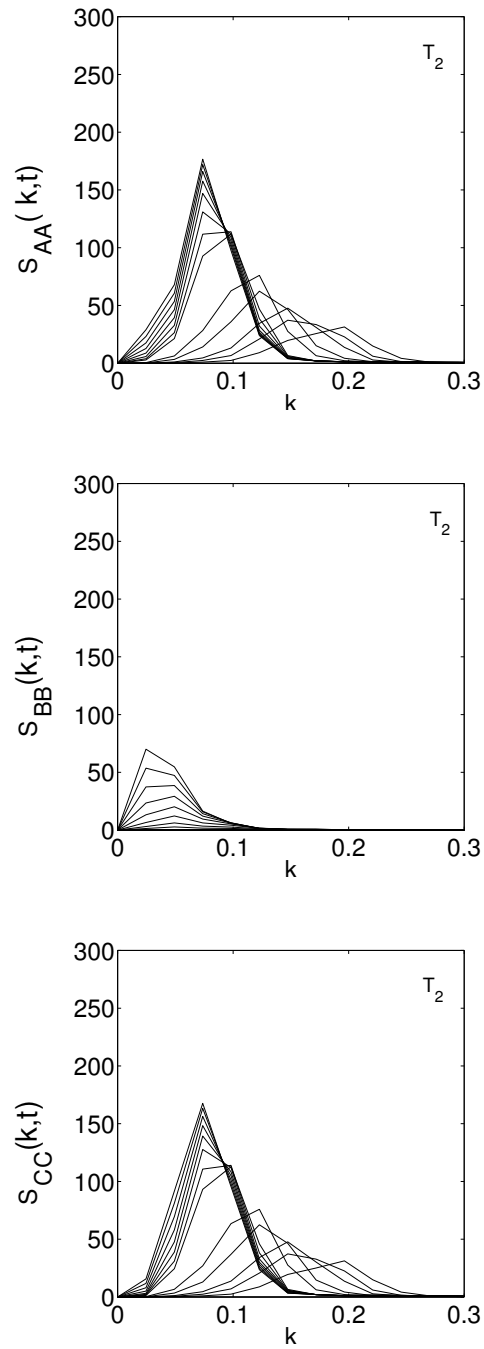


Figure 4.22: Structure functions S_{AA} , S_{BB} , S_{CC} plotted as a function of k for alloy T_2 . In each figure, the times are for $t=300, 400, 500, 1000, 2000, 2250, 2500, 2750, 3000, 3250, 3500, 3750$.

For the late stages of coarsening during a spinodal decomposition process, the growth exponent is found to be $n = 3$ [42, 44]. The same exponent is predicted by the well-known LSW theory [43] for the coarsening of precipitates.

The coarsening behaviour of the A-rich, B-rich and the C-rich domains, formed during decomposition is studied using a characteristic length scale representing the size of the domains. The kinetics of coarsening can be studied by monitoring the time dependence of a characteristic length scale of the microstructure. There are several ways of characterizing the typical length scales, R_i . We have used $R_i = 1/k_{1(i)}$, where $k_{1(i)}$ is the first moment of $S_{ii}(\mathbf{k}, t)$ ($i = A, B, C$), defined by:

$$k_{1(i)} = \frac{\sum k S_{ii}(\mathbf{k}, t)}{\sum S_{ii}(\mathbf{k}, t)} \quad (4.4)$$

For alloys P_1 and P_2 , $k_{1(i)}^{-3}$ vs. t plots are shown in Fig. 4.23. Comparison of length scales of the alloys across different times reveal that the B-rich domains are much coarser for P_2 than in P_1 . Further, the rate of coarsening of the B-rich domains is faster than in P_1 , as expected, because of the higher interfacial energy associated with the B-rich phase in P_2 .

The $k_{1(i)}^{-3}$ vs. t plots for alloys Q_1 , Q_2 and R_2 are shown in Fig. 4.24. Since the species B and C in alloy R_2 are equivalent to species C and B, respectively, in alloys Q_1 and Q_2 , we have compared the B-rich domains in Q_1 and Q_2 with the C-rich domains in R_2 and vice versa. The coarsening rates of A-rich domains in these alloys are similar. The B-rich domains in Q_2 , associated with a higher interfacial energy, show a coarsening rate much higher than the corresponding B-rich domains in Q_1 and C-rich domains in R_2 . Similarly, the B-rich phase in R_2 shows the highest coarsening rate among the three. It should also be noted that the B-rich phases in Q_2 and R_2 have a coarser domain size as

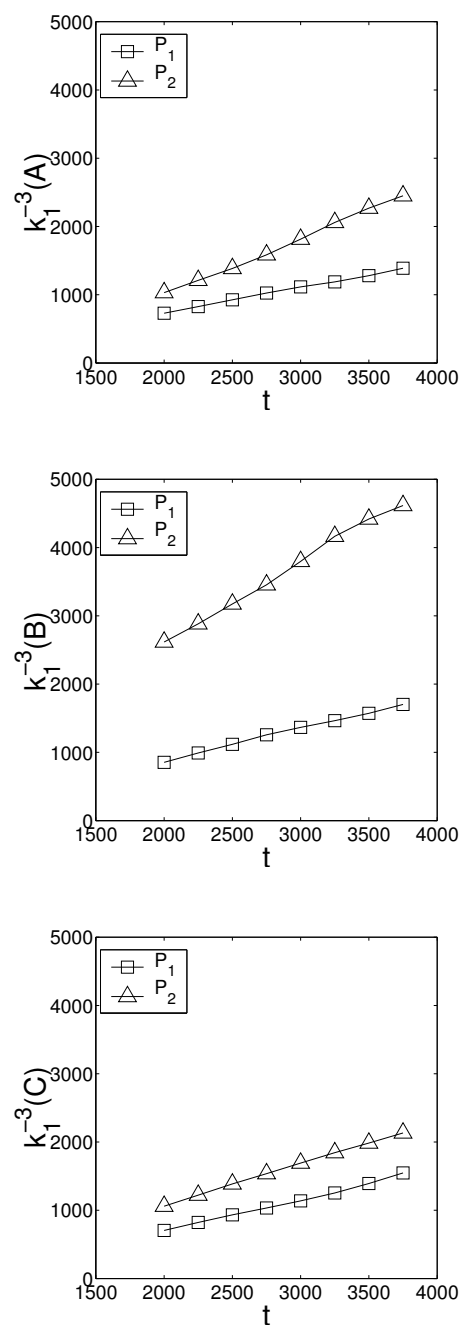


Figure 4.23: Plots of $k_{1(A)}^{-3}$, $k_{1(B)}^{-3}$, $k_{1(C)}^{-3}$ as a function of t for alloys P_1 and P_2 .

indicated in Fig. 4.24. Similarly, for alloys S_1 , S_2 and T_2 , (for which the plots are shown in Fig. 4.25), the rate of coarsening, as well as, the domain size of the B-rich domains in T_2 and S_2 is far larger than the A-rich and C-rich domains in these alloys.

So, it is evident from the study of coarsening kinetics that the β phase associated with higher interfacial energy coarsens more rapidly than those with lower energy.

4.8 Discussion

Effect of interfacial energy

The major conclusion that can be drawn from our study is that the morphology, as well as the kinetics of evolution, are significantly affected by the relative interfacial energies between the coexisting phases. Further, the effect depends on the alloy composition; more particularly, it depends on the concentration of the species (in our case, species B) associated with a higher interfacial energy. We illustrate this point using alloys R_2 and T_2 , and by comparing the microstructural evolution in them with that in the corresponding alloys Q_1 and S_1 , respectively.

In alloy R_2 , the composition of B is appreciable ($c_B = 0.5$); also, since it lies on the line Be in Fig. 4.2, the other two species, A and C, are equivalent. This implies that the initial decomposition necessarily involves the partitioning of species B, which is associated with a high interfacial energy. Thus, the initial decomposition is sluggish, and has a much higher wavelength, as compared to alloy Q_1 , its corresponding alloy in system 1. This ‘coarse’-ness of the microstructure is maintained till the very late stages.

In alloy T_2 , in contrast, the composition of B is very small ($c_B = 0.14$). Due to

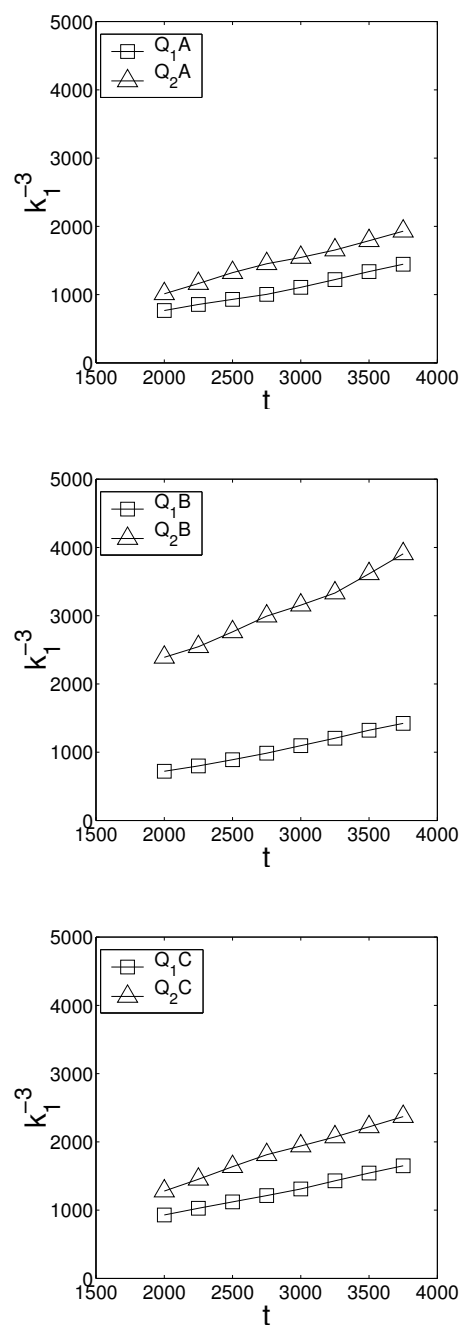


Figure 4.24: Plots of $k_{1(A)}^{-3}$, $k_{1(B)}^{-3}$, $k_{1(C)}^{-3}$ as a function of t for alloys Q_1 , Q_2 and R_2 . Species B and C in alloy Q_2 are equivalent to C and B, respectively, in alloy R_2 .

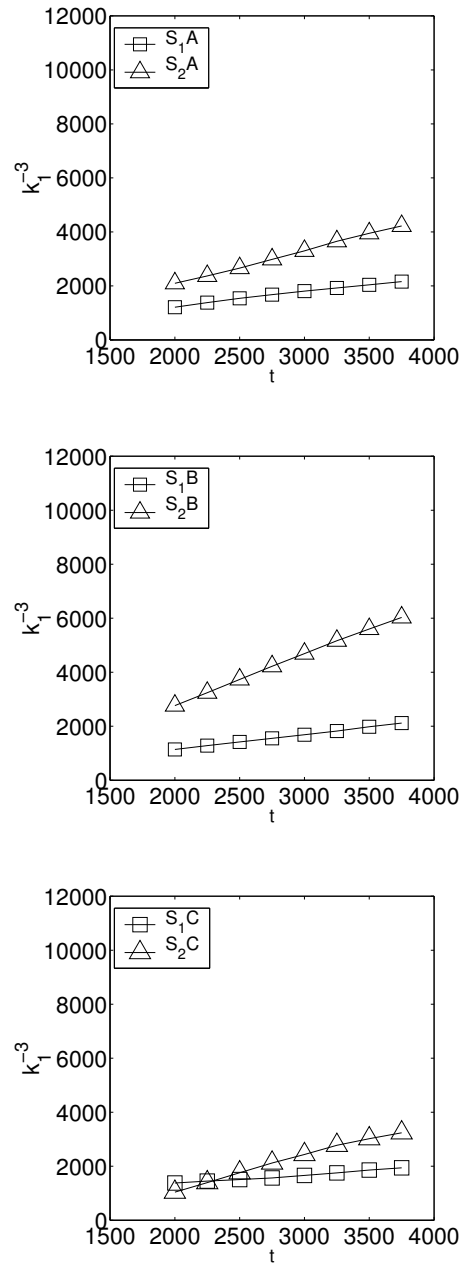


Figure 4.25: Plots of $k_{1(A)}^{-3}$, $k_{1(B)}^{-3}$, $k_{1(C)}^{-3}$ as a function of t for alloys S_1 , S_2 and T_2 . Species B and C in alloy S_2 are equivalent to C and B, respectively, in alloy T_2 .

the symmetry with respect to species A and C, and since the largest driving force for decomposition is along a line parallel to the AC axis, the initial separation does not involve partitioning of species B. Therefore, the length and time scales of early stage decomposition are similar in this alloy to those in alloy S_1 , the corresponding alloy in system 1 (The late appearance of B in a coarse form is again attributed to high $\sigma_{\alpha\beta}$ and $\sigma_{\beta\gamma}$ values).

The difference in the decomposition behaviour of these two alloys can be clearly attributed to whether species B needs to participate in early decomposition or not. In system 2, it depends essentially on the B content in the alloy. Thus, the five alloys we have studied may be grouped roughly into two categories: those with $c_B < 0.4$ do not require early-stage B-partitioning, and those with $c_B > 0.4$ require it. The foregoing illustration with alloys R_2 and T_2 shows that the interfacial energy effect is much greater in the second category of alloys than in the first.

Our system 1 is very similar to those studied by Chen [14, 32] and Eyre [15]; therefore, our microstructures for this system also closely resemble those obtained by them. By extending our study to system 2 (in which the interfacial energies are quite different), and comparing it with system 1, we have been able to identify the effect of interfacial energy on microstructural evolution. More importantly, we have been able to show that this effect is not uniform across all the alloy compositions, but depends essentially on the concentration of the high- σ species (in our case, species B) in the alloy.

Our results on alloys P_1 and P_2 (with equal A-, B- and C-content) have shown conclusively that differential interfacial energies do result in different composition history of each phase in the two systems. While such an analysis has not been presented explicitly in this thesis for the other alloys, this effect is implicit in the results on microstructure and

structure function.

Coarsening: is there dynamical scaling?

Our results on coarsening kinetics are interesting from two viewpoints. Our plots in Figs. 4.23- 4.25 may be taken to be evidence for the classical LSW temporal power law $R^3 \propto t$. However, such a conclusion is not warranted (particularly for system 2) for the following reason: The classical law is applicable for systems which are well into the dynamical scaling (DS) regime, in which the microstructures at different times are self-similar, and differ only in their length scales. However, in all the alloys in system 2, the coarsening rate of β particles is so much higher than that of either α or γ particles that the ratio of size scale of β to that of α (or, γ) phase keeps increasing with time. This observation clearly implies that the microstructures in all the alloys in system 2 are *not* self-similar, and hence the alloys are not in DS regime.

The coarsening rates of the three phases in each alloy in system 1 are similar; the structure functions, when scaled appropriately, still do not fall on a single curve (not shown here), as would be expected during DS. Thus, we conclude that the alloys in system 1 also do not exhibit DS.

It is possible that these alloys may enter a DS regime at a much later time. However, the restriction of the system size in our simulations to 256×256 does not allow us to access microstructures at times later than about $t > 4000$, by which the number of particles is already very low. Thus, longer simulations using a much larger system size would be very desirable to address the issue of dynamical scaling in ternary alloys.

Chapter 5

Conclusions

We have studied the role of interfacial energy on microstructural evolution when a homogeneous, disordered ternary alloy is quenched into a ternary miscibility gap. We have used computer simulations based on multicomponent Cahn-Hilliard equations for c_A and c_B , the compositions (in mole fraction) of A and B, respectively. The effect of interfacial energy (σ) is studied using two different systems: the first with equal σ 's (i.e. $\sigma_{\alpha\beta} = \sigma_{\beta\gamma} = \sigma_{\alpha\gamma}$), and the second with $\sigma_{\alpha\beta}$ and $\sigma_{\beta\gamma}$ which are about 2.2 times larger than $\sigma_{\alpha\gamma}$. We have chosen five different alloy compositions in these two systems.

System 1 is characterized by equal values of the interaction energies χ_{AB} , χ_{BC} and χ_{AC} (which determine the bulk thermodynamic behaviour of each of the three binary alloy systems); it is also characterized by equal values of the gradient energy coefficients, K_A , K_B and K_C (which determine the interfacial energies). Thus, it is an ideal system in which the three components are equivalent to one another. In our study, we use this system essentially for comparing our results from system 2 in which species B, since K_B is larger than $K_A = K_C$, is special (not equivalent to the other species).

In our detailed study of the five different alloys in the two systems, we have analyzed the interfacial energy effects in terms of microstructures, structure functions, kinetic paths and kinetics of coarsening. Our results indicate that the higher energy of the interfaces around the B-rich β phase has a different effect on two different categories of alloys: those (with $c_B < 0.4$) in which initial decomposition does not involve partitioning of species B, and those (with $c_B > 0.4$) in which it does. The interfacial energy effects are much stronger in the latter category of alloys than those in the former. For each category, the broad conclusions are listed as follows:

Category I: Alloys with $c_B < 0.4$ (P_2 , Q_2 and T_2):

In these alloys, early stage decomposition is characterized by a lack of partitioning of species B. Hence, the alloys undergo a pseudo-binary phase separation leading to the formation of A-rich and C-rich regions. The length scale of the initially decomposed domains in system 2 is larger than that in system 1 by about 10-15 %. Partitioning of B begins only during intermediate stages, and results in the formation of coarse β particles in the regions between the α and γ particles. The β particles in system 2 are larger, and coarsen faster, than those in system 1.

Category II. Alloys with $c_B > 0.4$ (R_2 and S_2):

In these alloys, partitioning of species B is essential right from the beginning of decomposition, and creates coarse B-rich and B-poor regions; due to the larger length scale of these microstructures, which are about 40-50 % larger than that in system 1, the initial decomposition is also much more sluggish than in system 1. This sluggishness is also evident in the time taken for the microstructure to exhibit all the three phases ($t \sim 1000$ in system 2, as opposed to $t \sim 500$ in system 1). Thus, the effect of interfacial

energy is much stronger in this category of alloys than those in Category I.

In all the alloys, the microstructures are coarser in system 2 than in system 1. This coarseness is more pronounced for the β particles than for the α and γ particles. Since the β phase has higher energy interfaces than the other two phases, its coarsening rate is much higher in system 2 than in system 1. However, there is not much difference in the coarsening rates of the other two phases across the two systems.

Bibliography

- [1] P. Haasen, *Physical metallurgy*, Cambridge University Press, 1997, 1.
- [2] J. E. Hilliard, *Phase Transformations*, (ed.), H. I. Aaronson, American Society for Metals, Metals Park, Ohio, 1970, Chapter 12, 497.
- [3] J. D. Gunton, M. San Miguel, and P. S. Sahni, *Phase Transitions and Critical Phenomena*, (ed.), C. Domb, and J. L. Lebowitz, Plenum Press, New York, 1983, **8**, 267.
- [4] K. Binder, *Materials science and technology*, (ed.), R. W. Cahn, P. Haasen, and E. J. Kramer, VCH, Weinheim, New York, 1991, 405.
- [5] Meijering, J. L., *Philips Res. Rep.*, 1950, **5**, 333.
- [6] Meijering, J. L., *Philips Res. Rep.*, 1951, **6**, 183.
- [7] R. Kikuchi, D. de Fontaine, M. Murakami, and T. Nakamura, *Acta. Metall.*, 1977, **25**, 207.
- [8] R. Kikuchi, *Acta. Metall.*, 1977, **25**, 195.
- [9] D. de Fontaine, *J. Phys. Chem. Solids*, 1972, **33**, 297.

- [10] D. de Fontaine, *J. Phys. Chem. Solids*, 1973, **34**, 1285.
- [11] J. E. Morral, and J. W. Cahn, *Acta Metall.*, 1971, **19**, 1037.
- [12] J. J. Hoyt, *Acta Metall.*, 1989, **37**, 2489.
- [13] J. S. Langer, *Ann. Phys.*, 1971, **65**, 53.
- [14] L. Q. Chen, *Acta. Metall. Mater.*, 1994, **42**, 3503.
- [15] D. J. Eyre, *SIAM J. Appl. Math.*, 1993, **53**, 6, 1686.
- [16] J. W. Cahn, and J. E. Hilliard, *J. Chem. Phys.*, 1958, **28**, 258.
- [17] K. B. Rundman, and J. E. Hilliard, *Acta Metall.*, 1966, **28**, 1025.
- [18] J. W. Cahn, *Acta Metall.*, 1961, **9**, 795.
- [19] D. de Fontaine, *Solid State Physics*, (ed.), H. Ehrenreich, F. Seitz, and D. Turnbull, Academic Press, 1979, **34**, 73.
- [20] R. J. Livak, and G. Thomas, *Acta Metall.*, 1974, **22**, 589.
- [21] E. P. Butler, and G. Thomas, *Acta Metall.*, 1970, **18**, 347.
- [22] C. K. Wu, and G. Thomas, *Met. Trans. A*, 1977, **8A**, 1911.
- [23] P. L. Makin, and B. Ralph, *J. Mat. Sci.*, 1984, **19**, 3835.
- [24] K. S. Vecchio, and D. B. Williams, *Acta Metall.*, 1987, **35**, 2959.
- [25] J. He, G. Han, S. Fukuyama, and K. Yokogawa, *Acta Mater.*, 1998, **46**, 215.
- [26] T. Shibita, Y. Shudo, and Y. Yoshino, *J. Japan Inst. Metals*, 1996, **60**, 802.

- [27] T. J. Cavanaugh, A. P. Russo, and E. B. Nauman, *Chemtech*, 1996, **26**, 8, 27.
- [28] R. Borsali, L. Giebel, E. W. Fischer, and G. Meier, *J. Non-cryst. Solids*, 1991, **130**, 2, 816.
- [29] D. Sutton, J. L. Stanford, and A. J. Ryan, *J. Macromol. Sci. B*, 2001, **40**, 3-4, 485.
- [30] J. S. Langer, *Acta Metall.*, 1973, **21**, 1649.
- [31] J. J. Hoyt, *Acta Metall.*, 1990, **38**, 227.
- [32] L. Q. Chen, *Scripta Metall. Mater.*, 1993, **29**, 683.
- [33] C. Huang, M. Olvera de la Cruz, and B. W. Swift, *Macromolecules*, 1995, **28**, 7996.
- [34] D. de Fontaine, *A computer simulation of the evolution of coherent composition variations in solid solutions*, Ph. D. Thesis, Northwestern University, Evanston, Illinois, 1967.
- [35] C. Huang, M. Olvera dela Cruz, and P. W. Voorhees, *Acta Mater.*, 1999, **47**, 4449.
- [36] P. G. Shewmon, *Diffusion in solids*, McGraw-Hill Book Company, Inc., New York, 1963.
- [37] J. W. Cahn, *Trans. Met. Soc. AMIE*, 1968, **242**, 166.
- [38] A. R. Allnatt, and A. B. Lidiard, *Atomic transport in solids*, Cambridge University Press, 1993.
- [39] L. Q. Chen, and J. Shen, *Comput. Phys. Commun.*, 1998, **108**, 147.
- [40] J. Zhu, L.Q. Chen, J. Shen and V. Tikare *Phys. Rev. E*, 1999, **60**,4, 3564.

- [41] M. Frigo and S.G.Johnson FFTW: An Adaptive Software Architecture For The FFT, ICASSP 98, 1998, **3**,1381 <http://www.fftw.org>
- [42] A. Chakrabarti, R. Toral, and J. D. Gunton, *Phys. Rev. E*, 1993, **47**, 5, 3025.
- [43] I. M. Lifshitz, and V. V. Slyozov, *J. Phys. Chem. Solids*, 1961, **19**, 35; C. Wagner, *Z. Elektrochem.*, 1961, **65**, 243.
- [44] T. M. Rogers, K. R. Elder, and R. C. Desai, *Phys. Rev. B*, 1988, **37**, 16, 9638.
- [45] C. Huang, and M. Olvera dela Cruz, *Macromolecules*, 1996, **29**, 6068.
- [46] C. Huang, and M. Olvera de la Cruz, *Europhys. Lett.*, 1996, **34**, 3, 171.
- [47] C. Huang, and M. Olvera de la Cruz, *Phys. Rev. E*, 1996, **53**, 1, 812.

Corrections to the ternary spinodal thesis

This reading copy corrects Eq. (3.17) on printed page 13 (PDF page 22). The original thesis remains available separately. Other original pages and results are unchanged. Correction date: 7 October 2026.

1. Cross mobility. The flux convention in Eq. (3.18) is $J_A = -M_{AA} \text{grad}(\mu_A - \mu_C) - M_{AB} \text{grad}(\mu_B - \mu_C)$, with the corresponding expression for B. Therefore the coefficient defined in Eq. (3.17) must have the opposite sign to its original printed expression. Equations (3.13) and (3.14) already contain the corresponding cross-flux term with the correct sign.

$$\begin{aligned} M_{AA} &= (1-c_A)^2 M_A + c_A^2 (M_B + M_C) \\ M_{BB} &= (1-c_B)^2 M_B + c_B^2 (M_A + M_C) \\ M_{AB} &= M_{BA} = -c_B(1-c_A)M_A - c_A(1-c_B)M_B + c_A c_B M_C. \end{aligned}$$

Here c_A , c_B and c_C are dimensionless mole fractions, with $c_A + c_B + c_C = 1$. M_A , M_B and M_C are the nonnegative component transport coefficients used in Eq. (3.5). The effective coefficients have the same units as those component coefficients; their physical units depend on the normalization of flux and chemical potential.

2. Thermodynamic consistency. The effective matrix is symmetric and positive semidefinite for nonnegative component coefficients. It can be written $R \text{diag}(M_A, M_B, M_C) R^T$, with rows $R = (1-c_A, -c_A, -c_A)$ and $(-c_B, 1-c_B, -c_B)$. It is positive definite when all three component coefficients are strictly positive. At positive absolute temperature, the flux contribution to entropy production is nonnegative. A cross coefficient need not be positive.

3. Two terminology and reference corrections. On synopsis page i, the matrix of second derivatives of bulk free energy is the Hessian, rather than the Jacobian of bulk free energy. On printed page 13, the sentence preceding Eq. (3.18) should refer to Eqs. (3.15)-(3.17), rather than Eq. (3.15) alone. These two corrections are recorded here; the original wording on those pages is retained.

This correction does not establish which expression was used in the historical simulation code. No historical simulations have been rerun for this correction, and no new numerical-validation claim is made.